Katalysys: k-ALM: IRRBB

IRRBB User Guide

25 November 2024 kalm@katalysys.com

Note: This document uses the term "interest" in alignment with UK regulatory guidance and market practice. For banks operating under Shariah principles, where interest (usury) is prohibited, the term “Interest” Rate Risk on the Banking Book (IRRBB) should be interpreted as "Profit” Rate Risk on the Banking Book (PRRBB). Similarly, any reference to "Net Interest Income" (NII) should be understood as "Net Income from Islamic Financing Transactions (NIIFT)."

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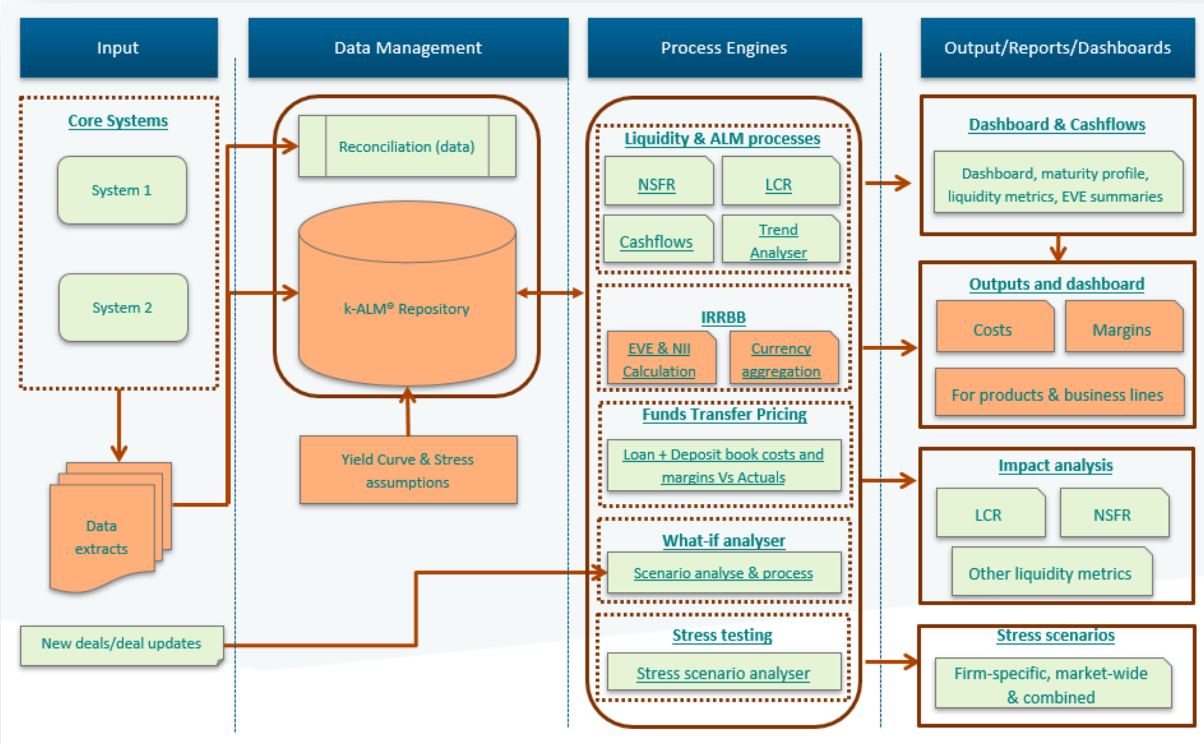
TECHNICAL COVERAGE

1 Introduction

k-ALM is a highly customisable software framework and toolkit developed by Katalysys. It can be used for stress testing, evaluating interest rate risk in the banking book (IRRBB) and for funds transfer pricing.

Data from the bank's various core banking systems is extracted, which is then combined with behavioural assumptions and interest rate curves which are collectively used to conduct the IRRBB measurements and assessments.

A high-level schematic diagram of the overall k-ALM toolkit is provided below:



(the elements for k-ALM IRRBB are highlighted in orange).

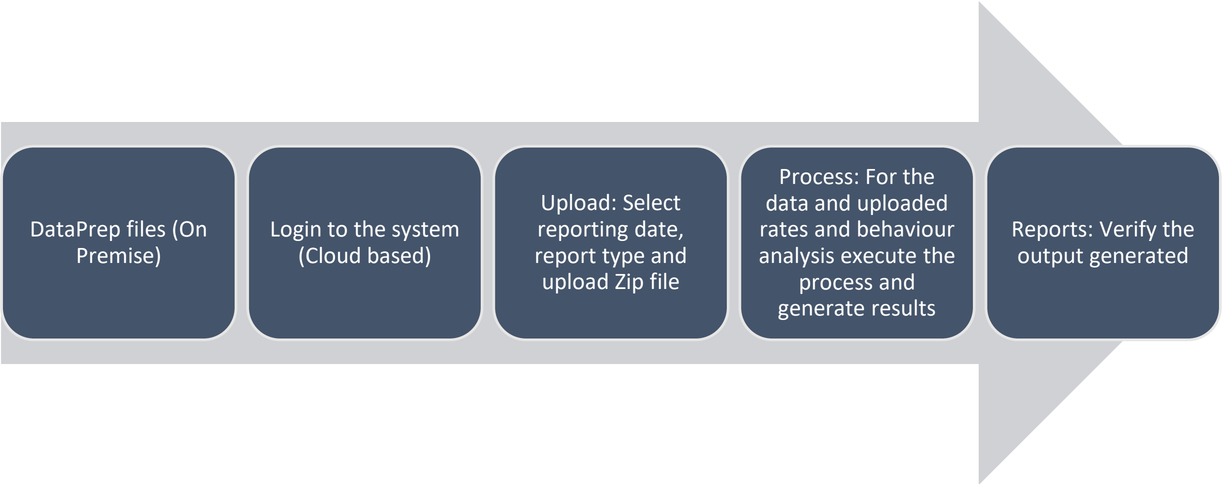
Throughout this document, a reference to IRRBB is equivalent to a reference to PRRBB, and any reference to interest is equivalent to a reference to profit.

2 Process Flow

The process to upload the data and generate reports and metrics is controlled via the k-ALM Application. There are two parts to this:

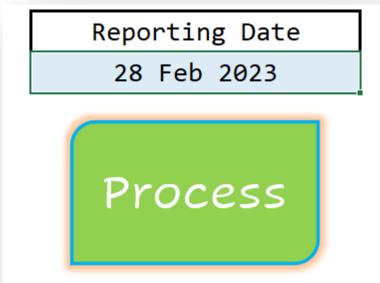
1) **ON PREMISE:** The desensitising of the data – which removes customer-sensitive information (e.g., customer identifier, account number, customer name).
2) **CLOUD:** Uploading and processing the data to generate reports and metrics.

The steps involved are shown below.



2.1 DataPrep Macro (On premise)

1) Ensure that all the data extract(s) are in the **DATA** folder.
2) Open the DataPrep Excel Macro (**kALM_BankId_DataPrepTool_n.n**[1]**.xlsm**)
3) Ensure direct entries and Parameters worksheets are up to date with the required data.
4) Enter the '**Reporting Date**' (blue shaded cell) and Click on the '**Process**' button



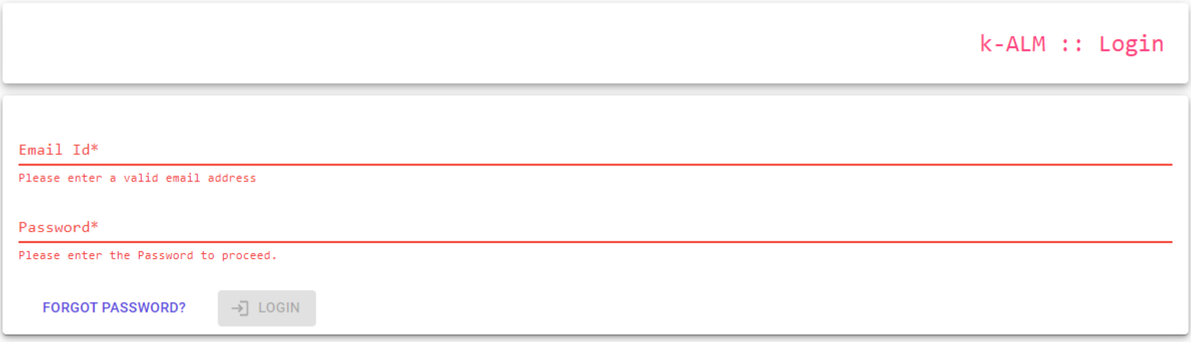
5) Once the Process is complete, a message is displayed.

The DataPrep Tool process flow is outlined in **Appendix D**.

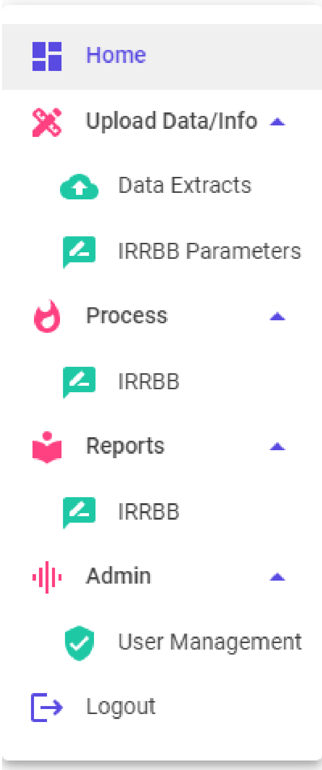
[1] Where n.n is the version number.

2.2 Executing the IRRBB Process (Cloud-based)

1) **Login** to the k-ALM Application [**https://bankname.k-alm.com/nnnnnnn[2]/Login**] with your user-id and password.

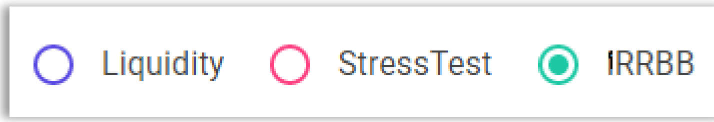


On the left side of the screen a navigation bar is provided.



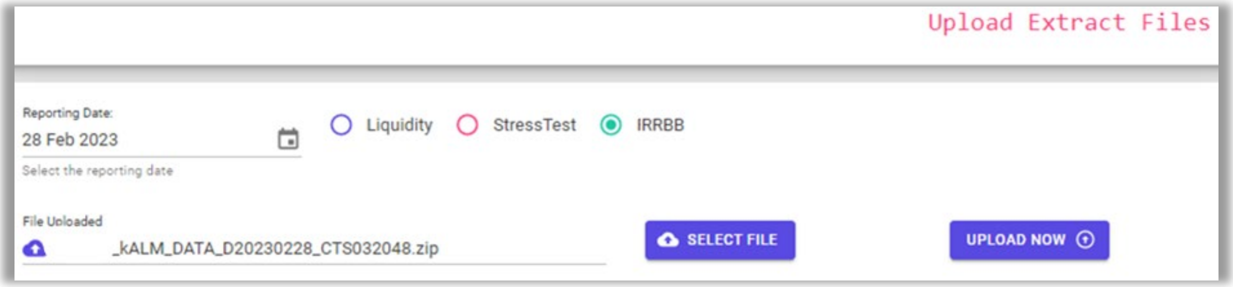
2) To begin the process, select **Data Extracts,** this is where the extract files can be uploaded for the specific Reporting Date.

3) Ensure that **IRRBB** is selected.



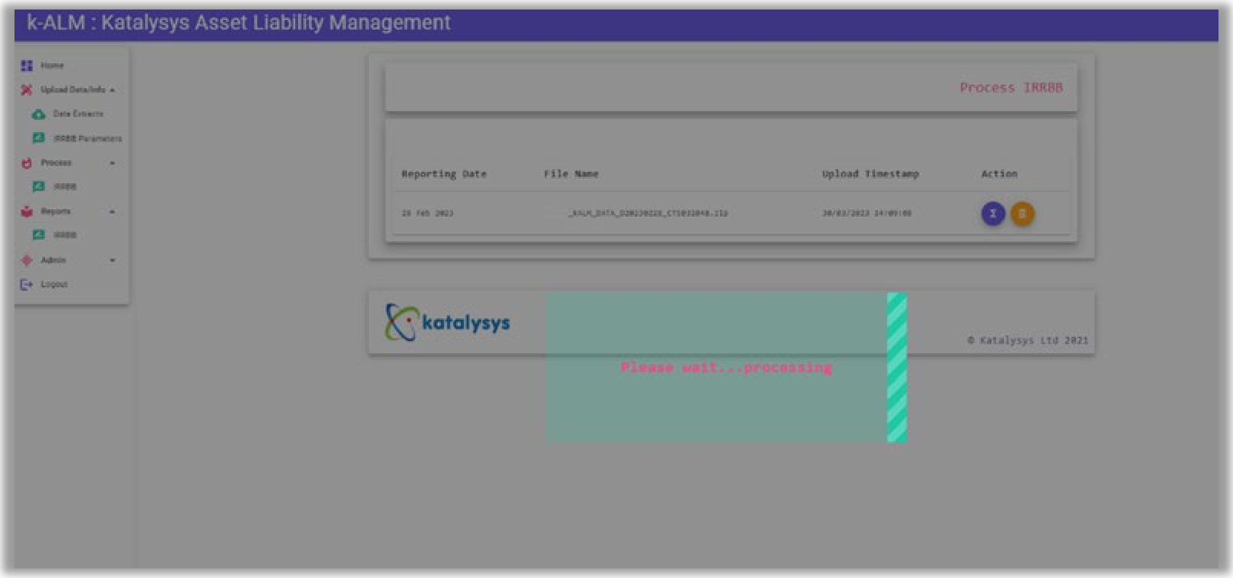
[2] Each bank is provided a unique id for enhanced security.

4) Select the **Reporting Date** and the corresponding Zip file (by clicking on 'SELECT FILES') to process and then click on 'UPLOAD NOW'.



5) The file that is uploaded is now ready for IRRBB mapping. Click on the **Purple button**, found under the Action column, as show below.

6) Once clicked, the process will execute, and the following screen will be shown.



7) Proceed to **Process → IRRBB**, identify the date for which the stress test has to be run and select the **Purple Arrow Button** under Action.

Please select the reporting date from the available dates shown below to calculate the as of IRRBB:

Reporting Date	Uploaded On	Processed On	Action
28 February 2023	29 Mar 2023 10:22:38	29 Mar 2023 10:22:38	›
31 January 2023	29 Mar 2023 10:12:52	29 Mar 2023 10:12:52	›

8) Select the correct rates to be used for the IRRBB calculation – proceed by clicking the **Purple Arrow Button** under the Action column. Directions to upload rate files can be found within Section 3.1.

Please select one (1) Rate set to be used for IRR Calculation.

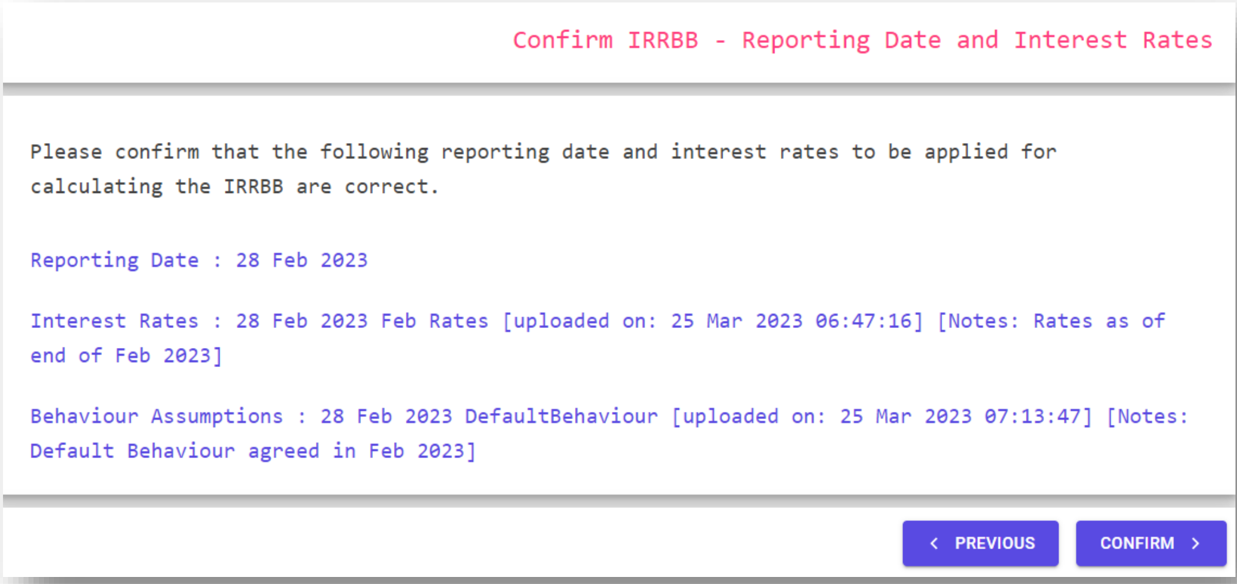
Reporting Date	Last Updated On	Notes	Action
28 Feb 2023	25 Mar 2023 06:47:16	Rates as of end of Feb 2023	›

9) Select the correct behaviour assumptions to be used for the IRRBB calculations – proceed by clicking the **Purple Arrow Button** under the Action column. Directions to upload behaviour files can be found within Section 3.1.

Please select one (1) set of Behaviour assumptions to be used for IRR Calculation.

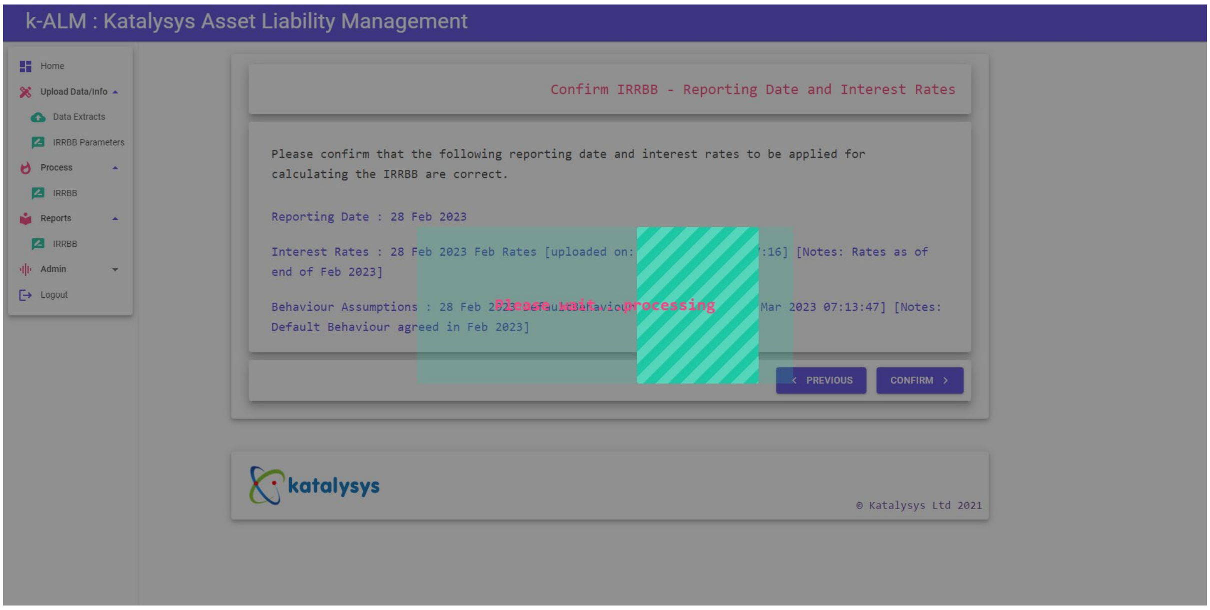
Reporting Date	Last Updated On	Notes	Action
28 Feb 2023	25 Mar 2023 07:13:47	Default Behaviour agreed in Feb 2023	›

10) Confirm the selection, ensuring the correct files are listed (files are shown for review by the user).
Once verified click **Confirm,** and the process will be executed.

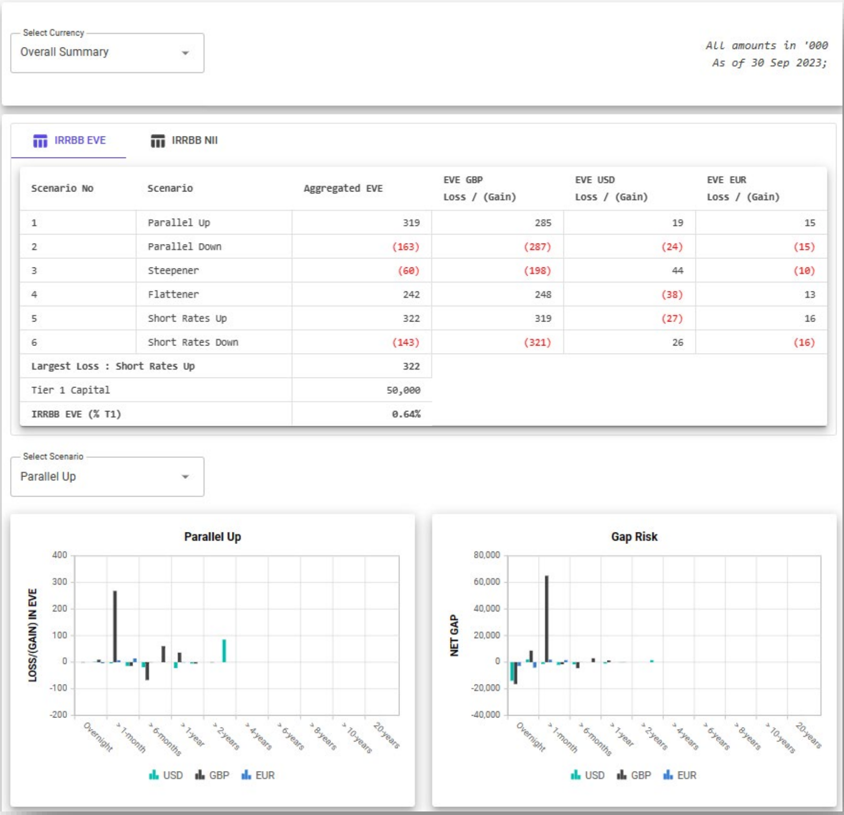


If any incorrect files have been selected, click the **Previous** Button to return to the desired page, and re-select the correct information (see Steps 8 and 9).

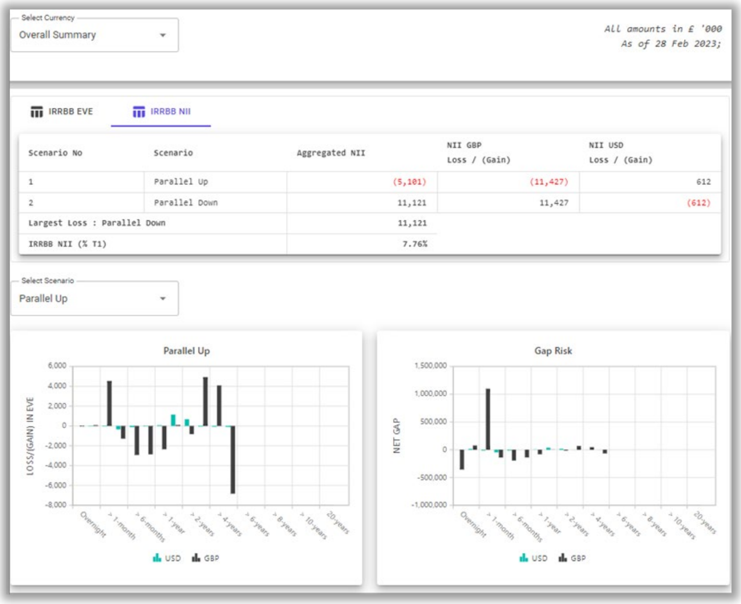
11) Once **Confirm** has been selected, the process will execute, and the following screen will be shown.



12) Upon completion of the process, the dashboard with the results are shown automatically, starting with overall summary. An example of the IRRBB EVE and IRRBB NII is shown below. The user can switch between the assessments (i.e., EVE or NII) by clicking on the tabs, located just above the initial results table.

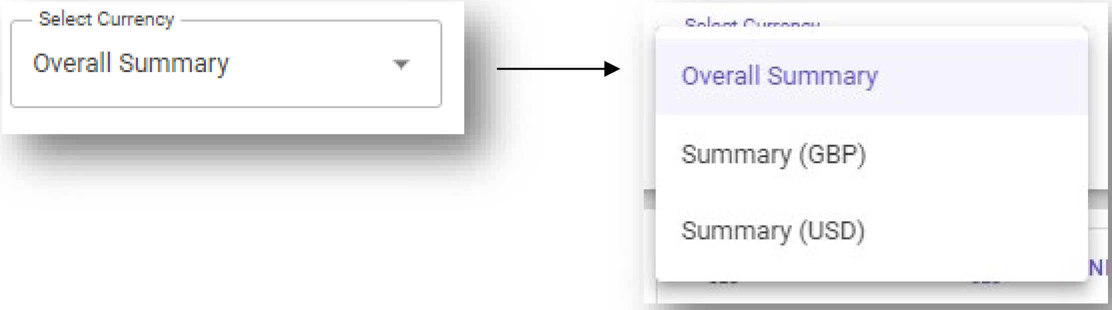


Scenario No	Scenario	Aggregated EVE	EVE GBP Loss / (Gain)	EVE USD Loss / (Gain)	EVE EUR Loss / (Gain)
1	Parallel Up	319	285	19	15
2	Parallel Down	(163)	(287)	(24)	(15)
3	Steepener	(60)	(198)	44	(10)
4	Flattener	242	248	(38)	13
5	Short Rates Up	322	319	(27)	16
6	Short Rates Down	(143)	(321)	26	(16)
Largest Loss : Short Rates Up		322			
Tier 1 Capital		50,000			
IRRBB EVE (% T1)		0.64%			

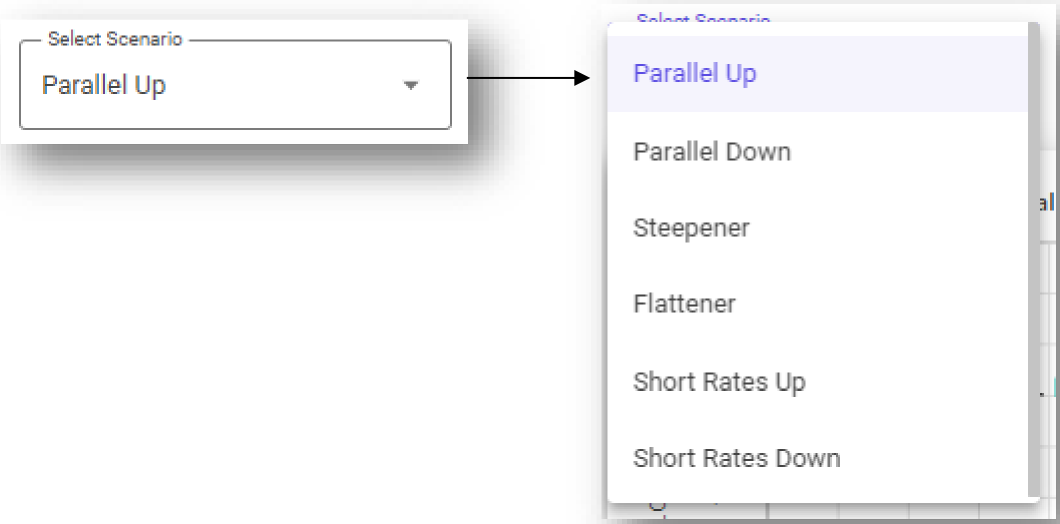


Scenario No	Scenario	Aggregated NII	NII GBP Loss / (Gain)	NII USD Loss / (Gain)
1	Parallel Up	(5,101)	(11,427)	612
2	Parallel Down	11,121	11,427	(612)
Largest Loss : Parallel Down		11,121		
IRRBB NII (% T1)		7.76%		

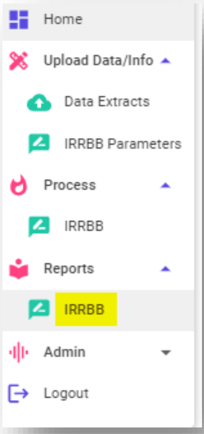
13) On this screen, the user can filter by currency using the dropdown list found in the top left of the summary page.



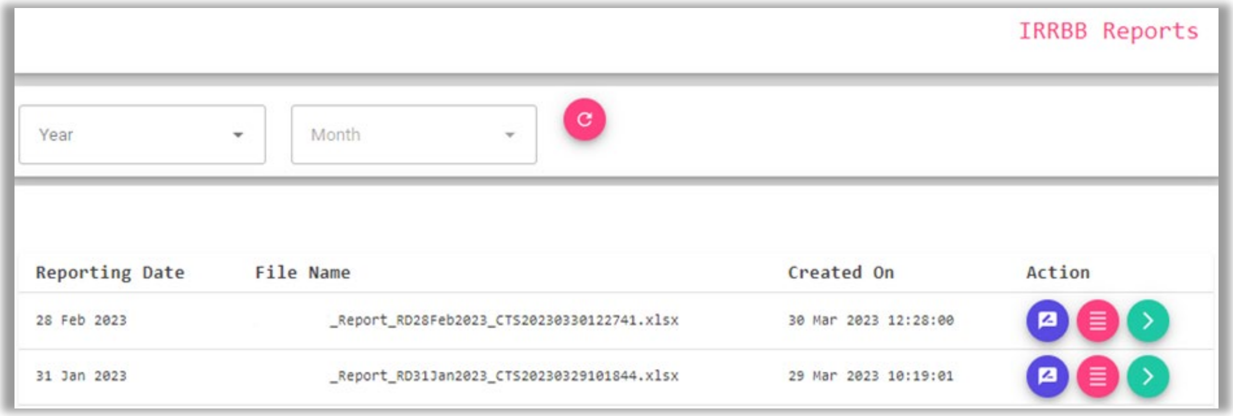
On this screen, the user can also filter by scenario using the dropdown list found towards the middle of the summary page (below the initial table).



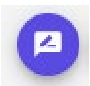
14) In order to allow further analysis and to support wider distribution, the full reports can be downloaded into Microsoft Excel. To do this, navigate to the navigation bar and click on **Report --> IRRBB.**



15) The following screen will be shown, providing all IRRBB reports.



Under the Action Column, the user can select one of the following options:



Purple Button – Click this to download the Microsoft Excel Report.

Pink Button – Click this to download the Microsoft Excel Datamaster (see section 3.2).

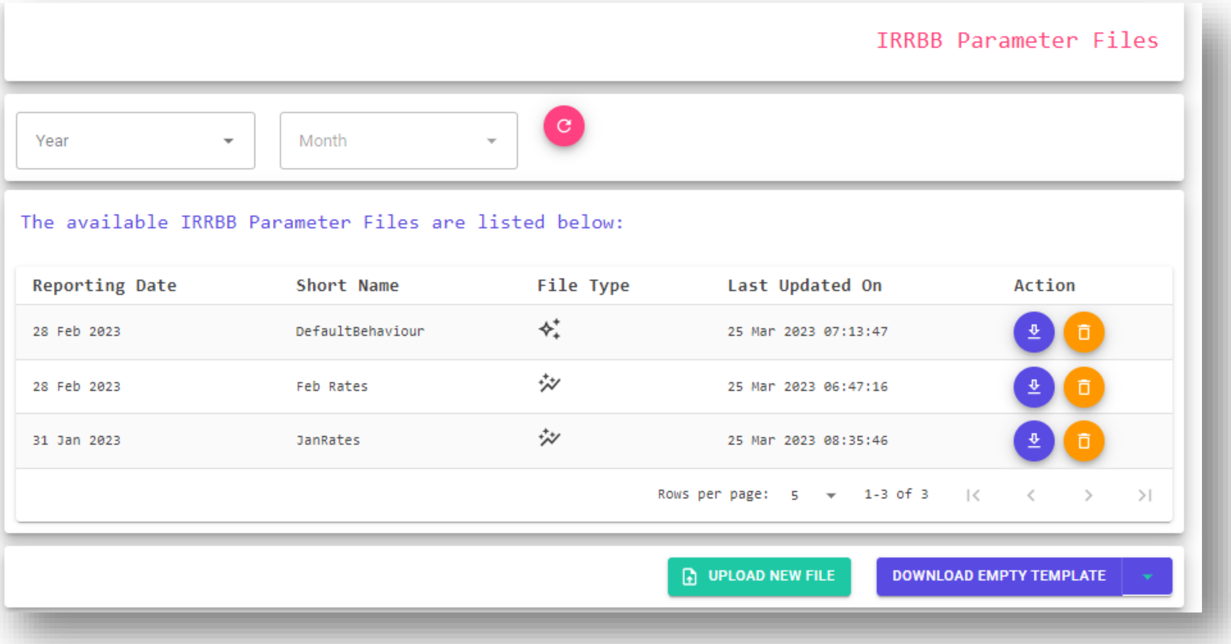
Green Button – Click this to re-access the dashboards on the k-ALM interface (as shown in step 12).

3 Additional functions

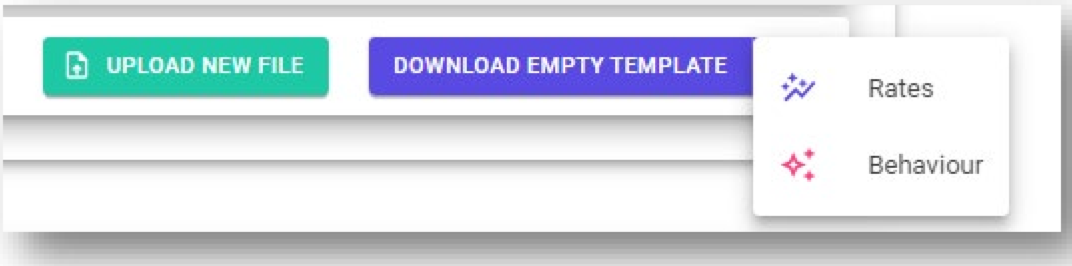
3.1 Uploading IRRBB Parameters – Behavioural Assumptions and Rates

The steps to upload a new set of rates or behavioural assumptions are given below.

1) On the left hand-side menu navigate to **Update Data/Info → IRRBB Parameters**



2) Click on 'DOWNLOAD EMPTY TEMPLATE' and select 'Rates' or 'Behaviour'.



Upon clicking the appropriate template, an empty excel template will be downloaded onto the user's local environment.

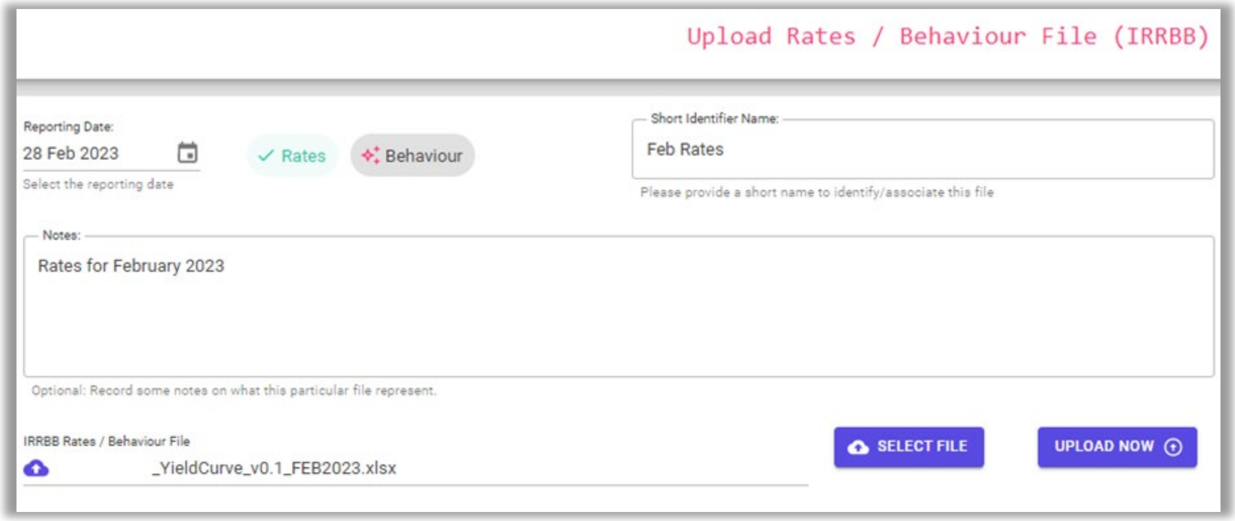
The files are as detailed below:

Rates	[XX]_[P/I]RRBB_YieldCurve_v0.1.xlsx
Behaviour	[XX]_[P/I]RRBB_BehaviourAdjustments_v0.1.xlsx

3) The user should then complete the relevant templates ensuring only the cells in **Blue** are updated.

4) Once templates are complete and ready, click on 'UPLOAD NEW FILE'

5) On the next screen, select the file type (Rates or Behaviour), enter a short identifier name (e.g., "Feb Rates") and add in any notes if required. Click on 'SELECT FILE' and navigate to the file. On completion, click 'UPLOAD NOW'.

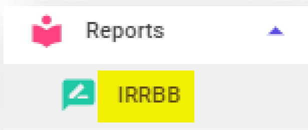


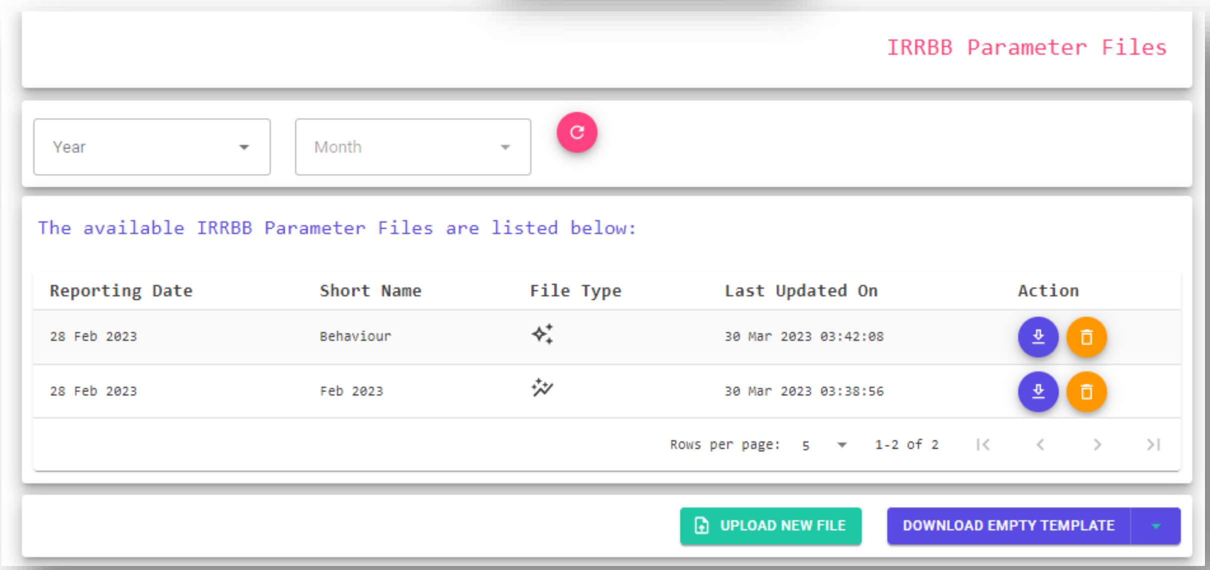
6) Once the upload is complete, the uploaded files will be shown, along with other existing uploads.

3.2 Detailed Data Download

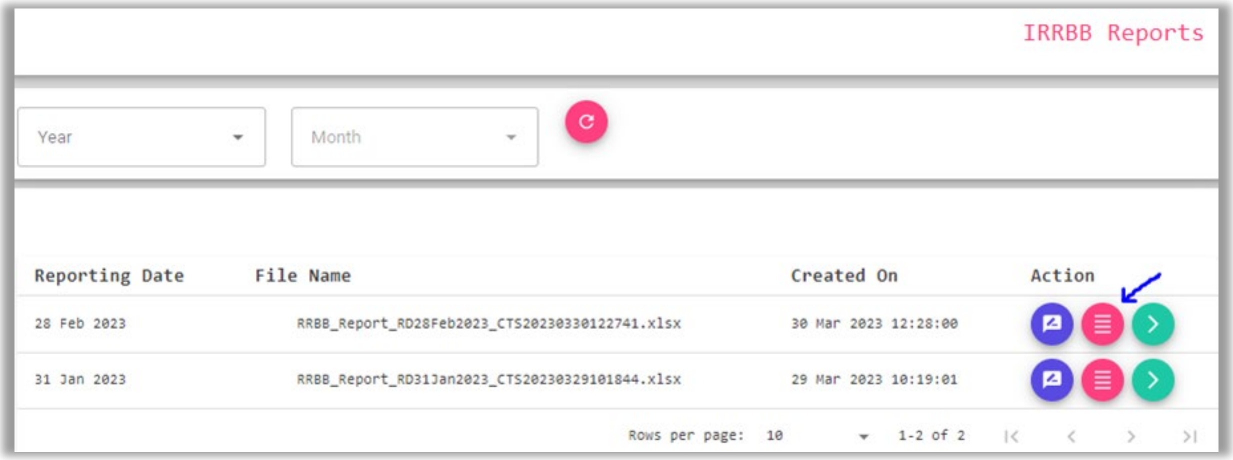
To download the detailed data of the IRRBB for a given date, use the following steps:

1) On the left-hand menu, navigate to **Reports → IRRBB**





2) Identify the appropriate reporting date and click on the **Pink button** under the Action Column (i.e., download datamaster).



Once completed, the datamaster will download on to the user's local machine, in Microsoft Excel format.

FUNCTIONAL COVERAGE

1 Introduction

This part of the document sets out the approach and inner workings of the k-ALM IRRBB Solution, which was developed by Katalysys and provides banks with a one-stop assessment tool to evaluate the EVE and NII impacts based on prescribed currency-specific interest rate shock scenarios, as well as the regulatory return template FSA017[3], in accordance with the guidance implemented by the PRA – based on Standards published by the Basel Committee on Banking Supervision (BCBS), which are set out in:

- the ICAA Part of the PRA Rulebook (ICAA Rules); and,
- SS31/15 'The Internal Capital Adequacy Assessment Process (ICAAP) and the Supervisory Review and Evaluation Process (SREP)' (SS31/15) (collectively, UK IRRBB Rules).

This tool and resulting outputs are designed to be practical and understandable by any user and to provide meaningful information for the risk management process that can be used as part of ongoing risk monitoring ad strategic risk decision making.

The IRRBB tool is designed to support senior management with the necessary information to make informed decisions regarding risk limits, risk policies and procedures, and the overall business strategy.

What is IRRBB?

IRRBB relates to present and future risks to a bank's capital and earnings arising from fluctuations in market interest rates.[4] In recent years, IRRBB has become an area of increased focus for regulators: this has coincided with significant changes in the interest rate environment across major economies, exiting a very long period of near-zero rates, heightened inflation, and industry events such as the failure of Silicon Valley Bank.

IRRBB can be sub-divided into components, which include:

- **Gap risk –** arising from mismatches in the re-pricing schedules of assets and liabilities.
- **Optionality risk** – arising from optionality embedded in contracts and from certain derivatives.
- **Basis risk** – arising from assets and liabilities being repriced against differing benchmarks.

At a basic level, IRRBB arises due to mismatches in the re-pricing schedules of assets and liabilities. IRRBB is typically quantified in two distinct ways.

1) **Economic Value of Equity measures**

   An EVE approach evaluates the change in the net present value of all cash flows originating from banking book assets, liabilities, and off-balance sheet items resulting from a change in interest rates and assuming a run-off balance sheet.

2) **Earnings-based measures**

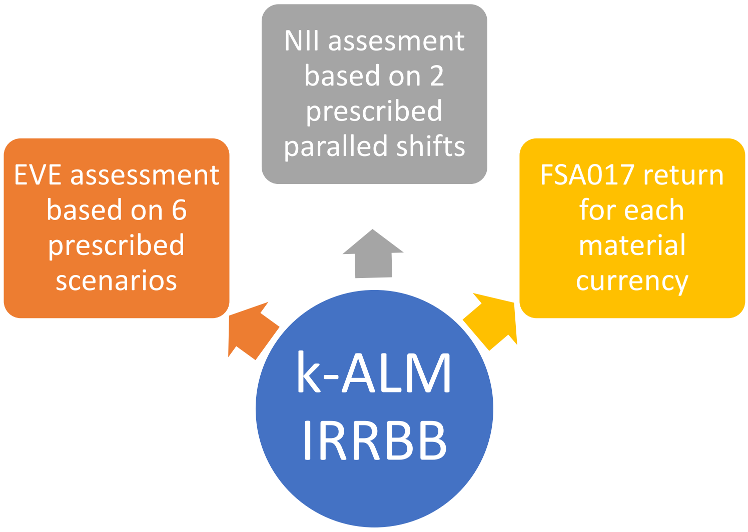
   An earnings-based approach evaluates the change in earnings (usually with a focus on net interest income) over a particular time horizon as a result of a movement in interest rates and based on certain balance sheet assumptions (e.g., run-off balance sheet, constant balance sheet, dynamic balance sheet).

[3] Only if subscribed separately, as this is an optional add-on.
[4] Basel Committee on Banking Supervision:Standards-Interest rate risk in the banking book,para. 8.

2 k-ALM IRRBB

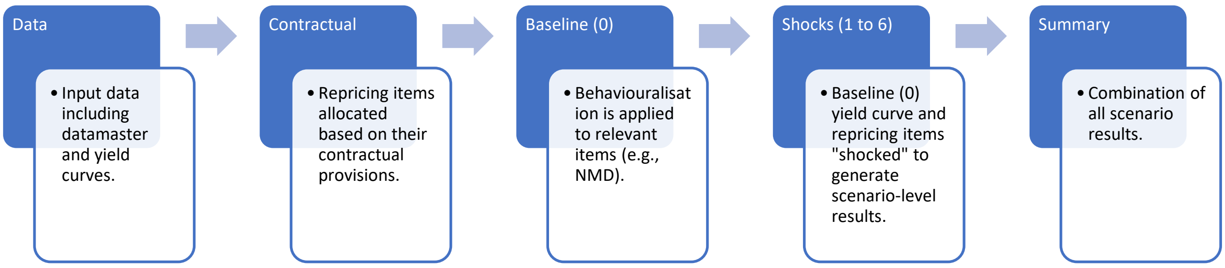
2.1 Objectives

The k-ALM IRRBB solution is designed to generate three fundamental outputs:



k-ALM IRRBB generates a detailed analysis of the Bank's EVE and NII assessments for each material currency at the given reference date. In addition, it will also allow the user to generate[5] an FSA017 return for each material currency in the prescribed format required by the PRA. Overall, these outputs will allow all key users to conduct an in-depth analysis and make informed decisions about interest risk management, limits, policies and procedures.

The overall process is achieved as follows:



2.2 Data Inputs

In order for the tool to produce such outputs, a number of inputs are required for the given reference date. A brief summary of the data required for each of the output are detailed below:

<u>IRRBB</u>

- Balance sheet data at the transaction level covering customer classification, product type and categorisation, currency, interest rate status, value, maturity date, and repricing date.

[5] Only if subscribed separately, as this is an optional add-on.

Behavioural Assumptions (if applicable)

- Split of non-maturity deposits by portfolio and each material currency between core and non-core elements, and a repricing time-bucket allocation of core deposits.
- The prepayment rate on retail fixed rate lending with prepayment risk, by portfolio and for each material currency.
- The redemption rate for retail fixed-term deposits with early termination risk, by portfolio and for each material currency.

Other

- Risk-free zero coupon yield curve for each material currency at prescribed tenors.
- Regulatory Tier 1 Capital value in reporting currency for the reference date.

2.3 Model Approach in line with Regulatory Framework

As detailed above, Banks are required to conduct an EVE assessment and NII assessment. This section will detail the granular approach that the tool undertakes to complete both assessments.

2.3.1 EVE Assessment

For many small- and medium-sized banks, the PRA allows for adoption of a Standardised Framework (SF) approach[6] which offers a clear and straight-forward route for measuring IRRBB under an EVE approach. The k-ALM tool follows this framework to ensure full compliance with the regulatory framework. The following narrative set out the application of the SF.

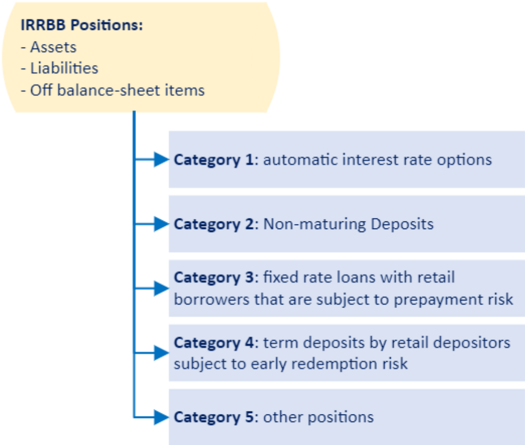
Material Currency Allocation

The Bank's material currencies (i.e. those currencies in which the firm has banking book assets (or liabilities) of more than 5% of total assets (or liabilities)) are identified, split by Assets, Liabilities and Off-balance sheet items[7], essentially curating a list of on/off balance sheet items by material currency.

Category Allocation and Sub-Allocation

[6] ICAA Rules, 9.13 to 9.43.
[7] Certain items are excluded e.g. Assets - CET1, fixed and intangible assets, and banking book equity exposures and Liabilities – CET1 capital.

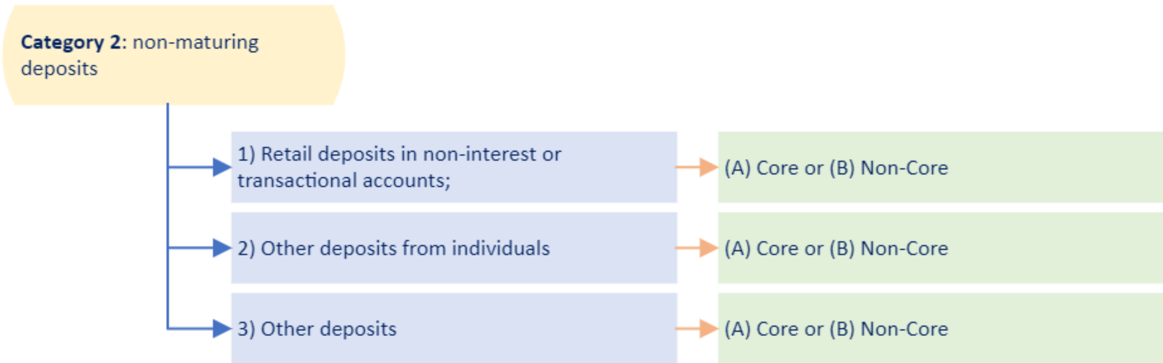
The items are then categorised into 5 high level categories:



For Categories 2, 3 & 4 these are then further allocated to sub-categories which are detailed below:

Category 2: Non-Maturing Deposits (NMD)

For NMDs, in each material currency, a further 2 levels of categorization is applied:

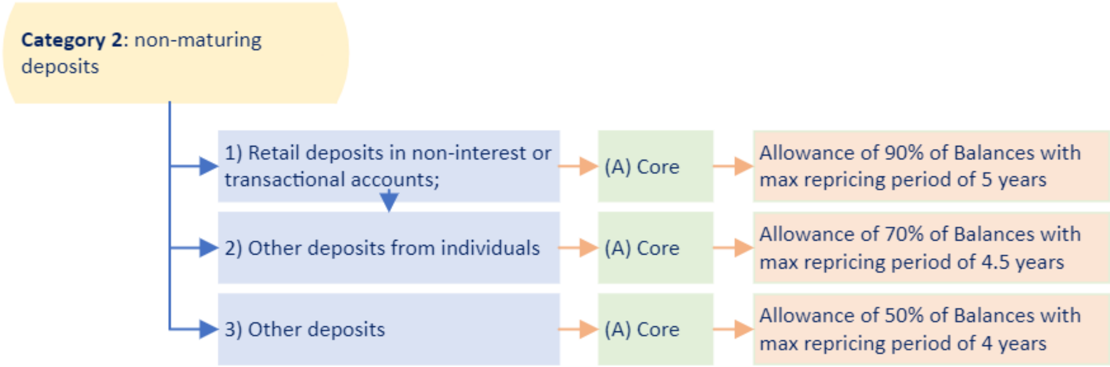


Core are those that are found to remain undrawn with a high degree of likelihood using data history of an appropriate length, and unlikely to re-price even under significant changes in the interest rate environment.

Non-Core represents any deposits that are not considered Core including any deposits from financial wholesale counterparties[8].

It should be noted that, certain limitations are required to be applied for certain categories, which include:

[8] SS31/15, 2.9B(vii)



Category 3: fixed rate loans with retail borrowers that are subject to prepayment risk

Items allocated by material currency within Category 3 are then categorised into 'homogeneous' portfolios. As suggested in the IRRBB Framework, 'homogeneous' portfolios are portfolios where the underlying assets are of similar nature[9]

For each portfolio, and in each material currency, a baseline Conditional Monthly Prepayment Rate (CMPR) is determined. A multiplier is then applied depending on the scenario. The multiplier, as set out in ICAA rules, are as follows:

Scenarios	Multiplier
1, 3, 5	0.8
2, 4, 6	1.2

Category 4: Term deposits by retail depositors subject to early redemption risk

Similar to approach undertaken for items allocated in Category 3. For each material currency items are categorised into 'homogeneous' portfolios.

For each portfolio, and in each material currency a baseline Term Deposit Redemption Ratio (TDRR) is determined. A multiplier is then applied depending on the scenario. The multiplier, as set out in ICAA rules, are as follows:

Scenarios	Multiplier
1, 4, 5	1.2
2, 3, 6	0.8

Re-pricing Cash flows Quantification

Upon completion of categorisation, for each category and sub-section and in each material currency – the reporting cashflows are determined including:

- Repayments of principal amounts;
- Re-pricing of principal amounts; and,
- Interest payment.

Where floating rate instruments are evident, the coupon payments up until the next re-pricing date is applied, along with the notional amount (next re-pricing date).

[9] E.g. Product type, Customer type, Geography, Sector etc.

Re-pricing Time Bucket Allocation

Each of the re-pricing cashflows, for each category and sub-allocation, and in each material currency - each item is allocated into a set of time buckets, as prescribed in the ICAA rules.

Risk Free Rates

Risk Free Rates are then applied to each re-pricing time bucket – such rates applied are open to the Bank's choice but as stated by the regulator, *"an appropriate general 'risk-free' yield curve per currency should be applied (e.g. swap rate curves). That curve should not include instrument-specific or entity-specific credit spreads or liquidity spreads"*[10]

Interest Rate Scenarios

Following the completion of items set above, the six pre-defined interest rate shift scenarios, for each material currency are set. These include:

Scenario	Description
1	Parallel up
2	Parallel down
3	Steepener
4	Flattener
5	Short rates up
6	Short rates down

The specific interest rate shifts by material currency also provided below:

Currency	Parallel	Short	Long
EUR	200	250	100
GBP	250	300	150
USD	200	300	150

A more detailed mapping of the SF to regulatory rules and applicability is provided in **Appendix A**.

[10] EBA/GL/2018/02, 115(n)

3 Dashboards

k-ALM IRRBB Dashboards are viewable while logged-in to the k-ALM Solution.

3.1 Overall Summary: EVE Results

The Overall Summary EVE Results table presents the outputs from each shock scenario on a consolidated currency basis. Individual currency results are combined in accordance with PRA Rules, which require that EVE losses are combined with EVE gains weighted using a factor of 50% (EVE $_{loss}$ x 100% + EVE $_{gain}$ x 50%). E.g., from the below, the result for the Short Rates Up scenario is: 319 x 100% + (27) x 50% + 16 x 100%= 322.

Result (gain)/loss

Currency level results

Scenario No	Scenario	Aggregated EVE	EVE GBP Loss / (Gain)	EVE USD Loss / (Gain)	EVE EUR Loss / (Gain)
1	Parallel Up	319	285	19	15
2	Parallel Down	(163)	(287)	(24)	(15)
3	Steepener	(60)	(198)	44	(10)
4	Flattener	242	248	(38)	13
5	Short Rates Up	322	319	(27)	16
6	Short Rates Down	(143)	(321)	26	(16)
Largest Loss : Short Rates Up		322			
Tier 1 Capital		50,000			
IRRBB EVE (% T1)		0.64%			

Result (gain)/loss

An EVE result indicates the change in NPV of all repricing asset and liability items between the baseline scenario (scenario 0) and the shock scenario (scenario 1 to 6). A negative result indicates an EVE $_{gain}$ whereas a positive result confirms an EVE $_{loss}$.

For risk evaluation, the largest EVE $_{loss}$ scenario is identified and expressed as a proportion of Tier 1 capital.

It is important to note that the EVE result may not be a true reflection of the accounting gain or loss that would result in the event of the coming to pass of the particular interest rate shock scenario.

3.2 Overall Summary: NII Results

The Overall Summary NII Results table presents the outputs from each shock scenario on a consolidated basis. Individual currency results are combined on the same basis as in the case of EVE, meaning that NII losses are combined with NII gains weighted using a factor of 50% (NII $_{loss}$ x 100% + NII $_{gain}$ x 50%). E.g., from the below, the result for the Parallel Down scenario is: 19,079 x 100% + (1,226) x 50% = 18,466.

IRRBB EVE | IRRBB NII

Scenario No	Scenario	Aggregated NII	NII GBP Loss / (Gain)	NII USD Loss / (Gain)
1	Parallel Up	(8,313)	(19,079)	1,226
2	Parallel Down	18,466	19,079	(1,226)
Largest Loss : Parallel Down		18,466		
IRRBB NII (% T1)		12.62%		

An NII result indicates the change in projected NII between the baseline scenario (scenario 0) and the shock scenario (scenario 1 and 2) assuming *inter alia* constant balance sheet and commercial margins. A negative result indicates an NII $_{gain}$ whereas a positive result confirms an NII $_{loss}$.

For risk evaluation, the largest NII $_{loss}$ scenario is identified and expressed as a proportion of Tier 1 capital.

3.3 Overall Summary: Charts

The right-hand chart shows the net gap by currency at each repricing time point. A positive net gap indicates an excess of repricing assets over repricing liabilities, and vice versa. Net gaps indicate the risk position at any particular repricing time bucket.

The left-hand chart presents a breakdown by currency of the EVE $_{(gain)/loss}$ at the level of the repricing time bucket. Users can interpret the relationship between the net gap and EVE result. Downward bars reflect an EVE $_{gain}$ at the particular bucket (i.e., an increase in the NPV of the net gap at the tenor point); conversely, upwards bars represent an EVE $_{loss}$ (i.e., a decrease in the NPV of the net gap at the tenor point).

The user can select the desired scenario from the drop-down menu, which will update the charts (since ΔEVE varies by scenario).

3.4 Currency Summary: EVE Results

The Currency Summary EVE Results table presents the outputs from each shock scenario on a currency basis.

IRRBB EVE | IRRBB NII

Parallel Up	285
Parallel Down	(287)
Steepener	(198)
Flattener	248
Short Rates Up	319
Short Rates Down	(321)
Largest Loss : Short Rates Up	319
Tier 1 Capital	50,000
IRRBB EVE (% T1)	0.64%

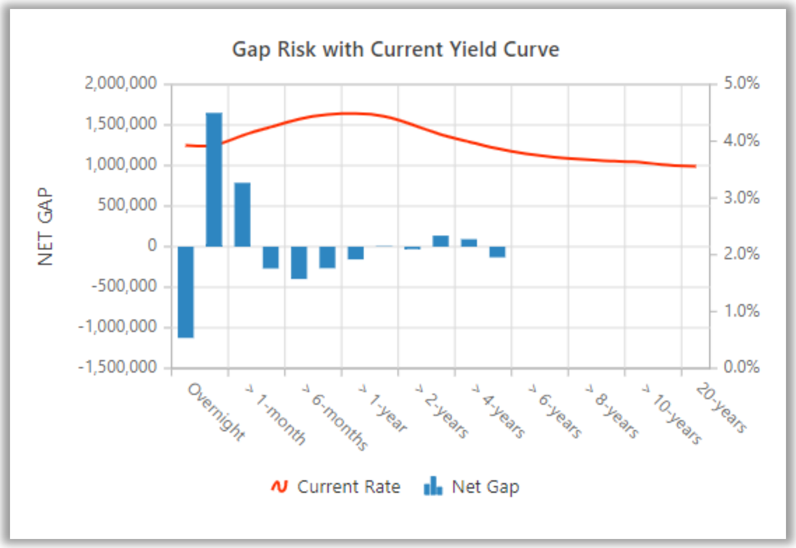
3.5 Currency Summary: NII Results

The Currency Summary NII Results table presents the outputs from each shock scenario on a currency basis.

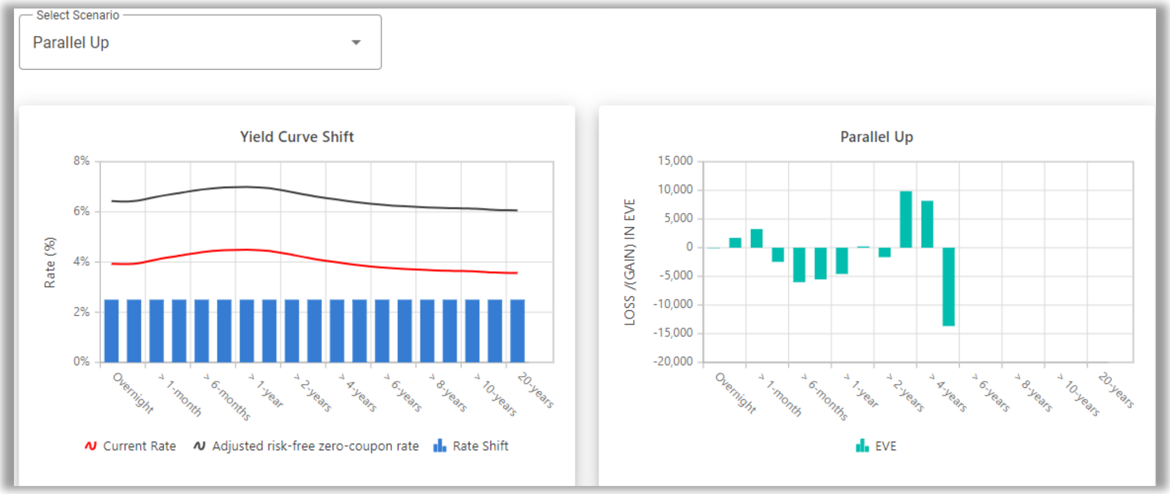
IRRBB EVE | IRRBB NII

Parallel Up	(19,079)
Parallel Down	19,079
Largest Loss : Parallel Down	19,079
IRRBB NII (% T1)	13.04%

3.6 Currency Summary: Charts



The top right chart shows the net gap at each repricing time point, overlaid with the input/baseline currency yield curve. A positive net gap indicates an excess of repricing assets over repricing liabilities, and vice versa. Net gaps indicate the risk position at any particular repricing time bucket. The input yield curve is adjusted in each scenario to generate IRRBB results.



In the bottom section, the user can select charts for each scenario from the drop-down menu.

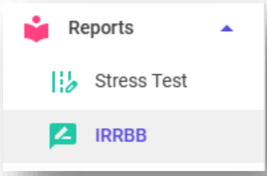
On the left-hand side, the scenario specific yield curves are shown:

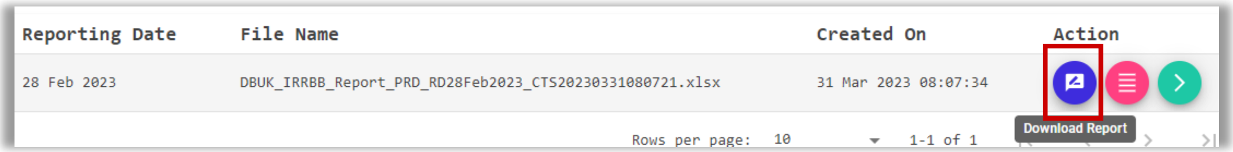
- Red line = input/baseline yield curve;
- Black line = shocked/adjusted yield curve; and,
- Blue bars = yield curve shift (black minus red) at each repricing time bucket.

On the right-hand side, the EVE chart presents a breakdown of the EVE $_{\text{(gain)/loss}}$ at the level of repricing time bucket.

4 Detailed Output Report

The k-ALM IRRBB outputs are downloadable in Microsoft Excel format from the Reports menu.





The output report is comprised on the following sections.

Sections	Output Section Names
Overall Summary	Table of results on a consolidated currency basis for EVE and NII measures. Consistent with the dashboards in the k-ALM user interface.
FSA017[11]	FSA017 return format and calculations in each currency. (available only under separate subscription)
Currency-level summary and calculations	Summary – table of results on a single currency basis for EVE and NII measures. Consistent with the dashboards in the k-ALM user interface.
	Contractual – repricing time bucket allocation of balance sheet items on a contractual basis.
	0_ to 6_ - baseline (0) and 6 shock scenario (1 to 6) calculations for the currency.
	NII – NII calculations for the currency.
	What-If analysis available in GBP[12].
Yield curves	Baseline and shocked yield curves in each material currency.
Inputs	Behavioural assumptions and baseline yield curves uploaded for IRRBB calculations (by currency).
	Datamaster and parameters.

4.1 Overall Summary

The Overall Summary mirrors the overall results as presented on the k-ALM IRRBB Dashboard (see Section 3). Here, both EVE and NII results are presented at individual currency level and combined under the required cross-currency consolidated method.

If provided, the risk appetite levels (R/A/G) are used to colour-code the results.

[11] Only if subscribed separately, as this is an optional add-on.
[12] Only if subscribed separately, as this is an optional add-on.

Interest/Profit Rate Risk in the Banking Book — Amounts in (£000's)

As of reporting date: 30-Sep-23

Scenario No	Scenario	Aggregated ΔEVE	% T1	ΔEVE GBP Loss / (Gain)	ΔEVE USD Loss / (Gain)	ΔEVE EUR Loss / (Gain)
1	Parallel Up	319	0.6%	285	19	15
2	Parallel Down	(163)	(0.3)%	(287)	(24)	(15)
3	Steepener	(60)	(0.1)%	(198)	44	(10)
4	Flattener	242	0.5%	248	(38)	13
5	Short Rates Up	322	0.6%	319	(27)	16
6	Short Rates Down	(143)	(0.3)%	(321)	26	(16)
Largest Loss: *Short Rates Up*		322				
Tier 1 Capital		50,000				
I/PRRBB EVE (% T1)		0.64%				

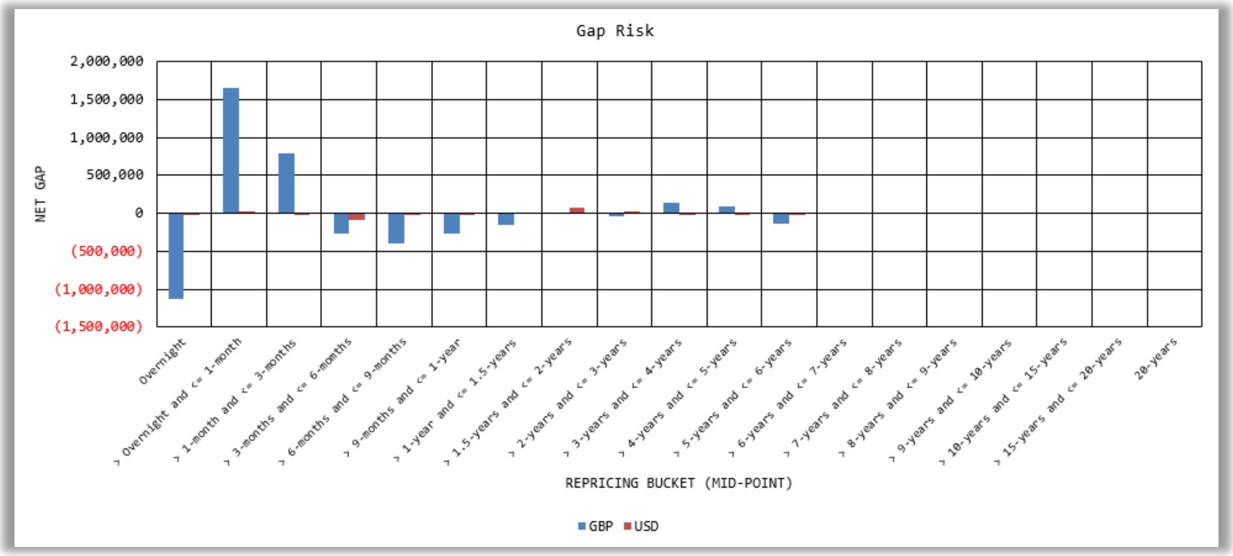
Scenario No	Scenario	ΔNII/NIIFT Loss / (Gain)	% T1	ΔNII/NIIFT GBP Loss / (Gain)	ΔNII/NIIFT USD Loss / (Gain)	ΔNII EUR Loss / (Gain)
1	Parallel Up	(165)	(0.3)%	(1,095)	299	83
2	Parallel Down	904	1.8%	1,095	(299)	(83)
Largest Loss: *Parallel Down*		904				
I/PRRBB NII/NIIFT (% T1)		1.81%				

Scenario:	Parallel Up

A uniform shift applied across the term structure, all rates are shifted upwards by the same amount.

FSA017 Results	
+200bp	116
-200bp	(123)

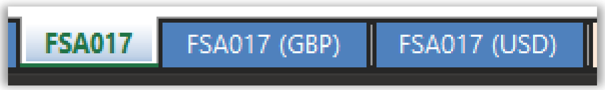
The user can also view similar charts, as required, by selecting from the drop-down menu in cell R6.



4.2 FSA017

This is an optional module, which is available via a separate subscription.

The FSA017 section references the underlying data and allocates repricing items to the required repricing buckets as specified by the FSA017 return template but using k-ALM IRRBB presentational convention. A calculation is performed in each of the material currencies, and on a consolidated basis (which, for FSA017, includes addition of non-material currencies).



The FSA017 core calculations are then re-presented in the formal return format:

FSA017
Interest rate gap report

		A	B	C	D	E	F
1	Gap sensitivity to parallel shift of:	2.00					
Assets		Totals	Overnight to 3 months	3-6 months	6-12 months	1-2 years	2-3 years
2	Monetary balance sheet assets (non-optional) as per contractual re-pricing date	4,608,462	3,300,523	150,817	100,304	341,129	217,489
3	> Adjustments for actual expected re-pricing date	0					
4	> Pipeline products	283,754	281,185	0	0	2,569	0
5	Monetary balance sheet assets with option features and with	0					
6	> Adjustments for actual expected re-pricing date	0					
7	> Pipeline products	0					
8	Net Trading Book asset	0					
9	Non-monetary and other assets	71,769					
10	Asset sub-total	4,963,986	3,581,709	150,817	100,304	343,698	217,489
11	> Linear derivative contracts	190	190	0	0	0	0
12	> Non-linear derivative contracts	0					
13	Asset totals	4,964,176	3,581,899	150,817	100,304	343,698	217,489

4.3 Currency-level Summary and Calculations

The Currency Summary and underlying Calculations detail the front-to-back process of generating IRRBB results at the material currency level.

The starting point is the contractual allocation of repricing balance sheet items.

Interest/Profit Rate Risk in the Banking Book - GBP (Amounts in £000s)	Bucket No:	1	2	3	4	
0 - Current Rates	Time Bucket:	Overnight	> Overnight and <= 1-month	> 1-month and <= 3-months	> 3-months and <= 6-months	> 6-mont and <= mont
28-Feb-23	Mid-Point (Yrs):	0.0028	0.0417	0.1667	0.375	0.6
Assets						
Category 1: Automatic Interest Rate Options	*0*	*0*	*0*	*0*	*0*	
Category 3: Retail Fixed Rate Loans with Prepayment Risk	*0*	*0*	*0*	*0*	*0*	
Category 5: Other Positions	*4,339,459*	*371*	*1,901,198*	*1,286,586*	*135,652*	*32,2*
Excluded items	*85,467*					
TOTAL ASSETS	4,424,926	371	1,901,198	1,286,586	135,652	32,2
Liabilities and Equity						

Contractual_GBP

After contractual allocation, behaviouralisation is applied as required to relevant balance sheet items, e.g., NMD, in tab "0_CCY". This sheet reflects the baseline scenario to which interest rate shocks are applied. Note that, for non-material currencies, their balance sheet items are presented on sheet "Contractual_Others", but are not further analysed for the six prescribed shock scenarios.

Interest/Profit Rate Risk in the Banking Book - GBP (Amounts in £000s)	Bucket No:	1	2	3	4	
0 - Current Rates	Time Bucket:	Overnight	> Overnight and <= 1-month	> 1-month and <= 3-months	> 3-months and <= 6-months	> 6-mont and <= mont
28-Feb-23	Mid-Point (Yrs):	0.0028	0.0417	0.1667	0.375	0.6
Assets						
Category 1: Automatic Interest Rate Options	*0*	*0*	*0*	*0*	*0*	
Category 3: Retail Fixed Rate Loans with Prepayment Risk	*0*	*0*	*0*	*0*	*0*	
Category 5: Other Positions	*4,339,459*	*371*	*1,901,198*	*1,286,586*	*135,652*	*32,2*
Excluded items	*85,467*					
TOTAL ASSETS	4,424,926	371	1,901,198	1,286,586	135,652	32,2
Liabilities and Equity						

0_GBP

The shock scenarios and resulting ΔEVE are detailed in scenario tabs "1_CCY" to "6_CCY".

Interest/Profit Rate Risk in the Banking Book - GBP (Amounts in £000s) 1 - Parallel Up 28-Feb-23	Bucket No: Time Bucket: Mid-Point (Yrs):	1 Overnight 0.0028	2 > Overnight and <= 1-month 0.0417	3 > 1-month and <= 3-months 0.1667
Assets				
Category 1: Automatic Interest Rate Options	*0*	*0*	*0*	*0*
Category 3: Retail Fixed Rate Loans with Prepayment Risk	*0*	*0*	*0*	*0*
Category 5: Other Positions	*4,339,459*	*371*	*1,901,198*	*1,286,586*
Excluded items	*85,467*			
TOTAL ASSETS	4,424,926	371	1,901,198	1,286,586
Liabilities and Equity				

Validation checks	
Check assets	0
Check liabilities and equity	0
Check off-balance sheet	0
Check net gap	0

1_GBP

These results are then collated and presented in the Summary tab for the currency.

Interest/Profit Rate Risk in the Banking Book - GBP   As of reporting date: 28-Feb-23

Scenario No	Scenario	ΔEVE Loss / (Gain)
1	Parallel Up	(10,773)
2	Parallel Down	11,381
3	Steepener	4,618
4	Flattener	(6,451)
5	Short Rates Up	(9,459)
6	Short Rates Down	9,642
Largest Loss:	*Parallel Down*	11,381
Tier 1 Capital		146,363
I/PRRBB EVE (% T1)		7.78%

Scenario:	Parallel Up

A uniform shift applied across the term structure, all rates are shifted upwards by the same amount.

PV01	44

Along with the EVE results for the currency, the k-ALM IRRBB Solution measures IRRBB on an NII basis. The calculations use the same allocations as EVE measures, and are detailed in tabs “CCY_NII”.

Interest/Profit Rate Risk in the Banking Book - GBP (Amounts in £000s)
1 - Parallel Up
28-Feb-23

Bucket No	Time Bucket	Mid-Point (Yrs)	Net Gap 0	Net Gap 1	Current Rates	Parallel Up	Rate Shift	ΔNII/NIIFT Loss/(Gain)
1	Overnight	0.0028	(1,125,320)	(1,125,320)	3.93%	6.43%	2.50%	28,054
2	> Overnight and <= 1-month	0.0417	1,649,007	1,649,007	3.93%	6.43%	2.50%	(39,506)
3	> 1-month and <= 3-months	0.1667	786,905	786,905	4.10%	6.60%	2.50%	(16,393)
4	> 3-months and <= 6-momths	0.375	(268,836)	(268,836)	4.25%	6.75%	2.50%	4,201
5	> 6-months and <= 9-months	0.625	(398,505)	(398,505)	4.39%	6.89%	2.50%	3,736
6	> 9-months and <= 1-year	0.875	(265,342)	(265,342)	4.47%	6.97%	2.50%	829
7	> 1-year and <= 1.5-years	1.25	(157,182)	(157,182)	4.49%	6.99%	2.50%	0
8	> 1.5-years and <= 2-years	1.75	5,165	5,165	4.44%	6.94%	2.50%	0
9	> 2-years and <= 3-years	2.5	(30,299)	(30,299)	4.29%	6.79%	2.50%	0
10	> 3-years and <= 4-years	3.5	135,791	135,791	4.12%	6.62%	2.50%	0
11	> 4-years and <= 5-years	4.5	91,903	91,903	3.99%	6.49%	2.50%	0
12	> 5-years and <= 6-years	5.5	(131,586)	(131,586)	3.87%	6.37%	2.50%	0
13	> 6-years and <= 7-years	6.5	0	0	3.78%	6.28%	2.50%	0
14	> 7-years and <= 8-years	7.5	0	0	3.72%	6.22%	2.50%	0
15	> 8-years and <= 9-years	8.5	0	0	3.68%	6.18%	2.50%	0
16	> 9-years and <= 10-years	9.5	0	0	3.65%	6.15%	2.50%	0
17	> 10-years and <= 15-years	12.5	0	0	3.63%	6.13%	2.50%	0
18	> 15-years and <= 20-years	17.5	0	0	3.58%	6.08%	2.50%	0
19	20-years	25	0	0	3.56%	6.06%	2.50%	0
20	No Specific Repricing		(291,687)	(291,687)				0
						Parallel Up	ΔNII/NIIFT Loss/(Gain)	(19,079)

GBP_NII-NIIFT

Returning to the Currency Summary, the NII results are shown below those for EVE.

Scenario No	Scenario	ΔNII/NIIFT Loss / (Gain)
1	Parallel Up	(19,079)
2	Parallel Down	19,079
Largest Loss: *Parallel Down*		19,079
I/PRRBB NII/NIIFT (% T1)		13.04%

The Currency Summary also presents charts in line with the same summaries available via the k-ALM IRRBB Dashboards.

4.3.1 What-if analysis

This is an optional module, which is available via a separate subscription.

Based on the core EVE and NII results, the user can undertake some What-If analysis in the primary currency, Sterling.

What-If (GBP)

The tab has 3 distinct what-if functions. In all cases, the user can adjust the purple cells to make the assessments under different parameters.

A) Estimate of the maximum capacity of additional assets and/or liabilities that can be undertaken based on the current rate environment, I/PRRBB profile, and selected scenario

Asset tenor	> 4-years and <= 5-years	*the repricing tenor of prospective assets*
Liability tenor	> 3-months and <= 6-momths	*the repricing tenor of liabilities to be generated, or assets to be replaced*
Scenario	Parallel Up	*NB - current maximum loss scenario is: SHORT RATES DOWN*
Rate shift - assets (bps)	250	*the rate shifts applied at each tenor based on the selected scenario*
Rate shift - liabilities (bps)	250	
Buffer (bps)	100	*the desired headroom to the EVE Amber limit (in basis points)*
Estimated Capacity (£'000s)	180,179	*the estimated capacity to take EVE I/PRRBB % Tier 1 Capital to the Amber limit (less any buffer applied)*
EVE impact loss/(gain) (£'000s)	14,533	*the EVE impact based on the selected parameters and estimated capacity*
NII/NIIFT impact loss/(gain) (£'000s)	2,815	*the value of the impact to NII/NIIFT assuming the above outputs*
% Tier 1 capital	5.4%	*resultant NII/NIIFT % Tier 1 Capital assuming the above outputs*

First is an estimate of the maximum capacity of additional assets and/or liabilities that can be undertaken based on the current rate environment, I/PRRBB profile, and selected scenario.

B) Estimate of the maximum capacity of additional assets and/or liabilities that can be undertaken based on the current rate environment, I/PRRBB profile, and selected scenario, with the added functionality to adjust Tier 1 Capital (e.g., to account for unaudited profits)

Asset tenor	> 4-years and <= 5-years	the repricing tenor of prospective assets
Liability tenor	> 3-months and <= 6-momths	the repricing tenor of liabilities to be generated, or assets to be replaced
Scenario	Parallel Up	NB - current maximum loss scenario is: SHORT RATES DOWN
Rate shift - assets (bps)	250	the rate shifts applied at each tenor based on the selected scenario
Rate shift - liabilities (bps)	250	
Buffer (bps)	100	the desired headroom to the EVE Amber limit (in basis points)
Tier 1 capital adjustment (£'000s)	10,000	the increase/(decrease) in tier 1 capital to be incorporated
Estimated Capacity (£'000s)	188,857	the estimated capacity to take EVE I/PRRBB % Tier 1 Capital to the Amber limit (less any buffer applied)
EVE impact loss/(gain) (£'000s)	15,233	the EVE impact based on the selected parameters, estimated capacity, and capital adjustment
NII/NIIFT impact loss/(gain) (£'000s)	2,951	the value of the impact to NII/NIIFT assuming the above outputs
% Tier 1 capital	4.9%	resultant NII/NIIFT % Tier 1 Capital assuming the above outputs

Second, is an estimate of the maximum capacity of additional assets and/or liabilities that can be undertaken based on the current rate environment, I/PRRBB profile, and selected scenario plus the addition of a Tier 1 capital adjustment (by increasing or decreasing the Tier 1 capital value, the available capacity changes compared to scenario A).

C) Estimated EVE impact (and resultant % Tier 1 Capital value) of additional assets and/or liabilities based on the current rate environment, I/PRRBB profile, and existing maximum loss scenario

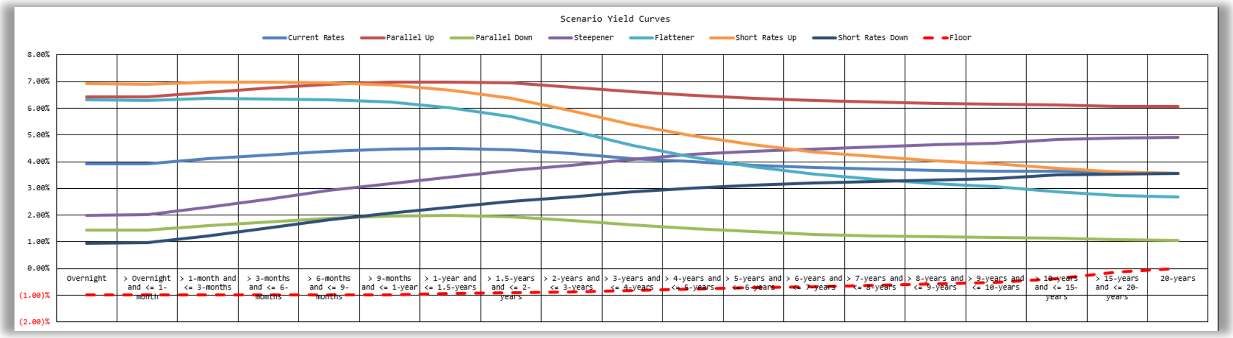
Asset tenor	> 1-month and <= 3-months	the intended tenor of prospective assets
Liability tenor	Overnight	the repricing tenor of liabilities to be generated, or assets to be replaced
Scenario	Parallel Down	NB - current maximum loss scenario is: SHORT RATES DOWN
Amount (£'000s)	50,000	the amount of assets and, or liabilities to factor into the calculation
Rate shift - assets (bps)	(250)	the rate shifts applied at each tenor based on the selected scenario
Rate shift - liabilities (bps)	(250)	
EVE impact Loss/(Gain) (£'000s)	(207)	the gross impact to EVE at the currency level
% Tier 1 capital	3.2%	resultant EVE % Tier 1 Capital assuming the above outputs
NII/NIIFT impact loss/(gain) (£'000s)	205	the value of the impact to NII/NIIFT assuming the above outputs
% Tier 1 capital	7.2%	resultant NII/NIIFT % Tier 1 Capital assuming the above outputs

Thirdly, an estimate of the EVE and NII impacts (and resultant % Tier 1 capital) of an adjustable asset/liability value and based on the current rate environment, I/PRRBB profile.

4.4 Yield Curves

This section provides the user with a table and chart of both the input yield curves, by currency, and the adjusted or shocked yield curves (as required by PRA Rules).

Bucket No	Time Bucket	Mid-Point (Yrs)	Current Rates 0	Parallel Up 1	Parallel Down 2	Steepener 3	Flattener 4	Short Rates Up 5	Short Rates Down 6	Floor
1	Overnight	0.0028	3.93%	6.43%	1.43%	1.98%	6.33%	6.93%	0.93%	(1.00)%
2	> Overnight and <= 1-month	0.0417	3.93%	6.43%	1.43%	2.01%	6.30%	6.90%	0.96%	(1.00)%
3	> 1-month and <= 3-months	0.1667	4.10%	6.60%	1.60%	2.28%	6.37%	6.98%	1.22%	(1.00)%
4	> 3-months and <= 6-momths	0.375	4.25%	6.75%	1.75%	2.60%	6.35%	6.98%	1.52%	(1.00)%
5	> 6-months and <= 9-months	0.625	4.39%	6.89%	1.89%	2.92%	6.31%	6.96%	1.82%	(1.00)%
6	> 9-months and <= 1-year	0.875	4.47%	6.97%	1.97%	3.17%	6.22%	6.88%	2.06%	(1.00)%
7	> 1-year and <= 1.5-years	1.25	4.49%	6.99%	1.99%	3.43%	6.00%	6.68%	2.30%	(0.94)%
8	> 1.5-years and <= 2-years	1.75	4.44%	6.94%	1.94%	3.66%	5.67%	6.38%	2.50%	(0.91)%
9	> 2-years and <= 3-years	2.5	4.29%	6.79%	1.79%	3.87%	5.16%	5.90%	2.68%	(0.88)%
10	> 3-years and <= 4-years	3.5	4.12%	6.62%	1.62%	4.09%	4.60%	5.37%	2.87%	(0.83)%
11	> 4-years and <= 5-years	4.5	3.99%	6.49%	1.49%	4.27%	4.16%	4.96%	3.02%	(0.78)%
12	> 5-years and <= 6-years	5.5	3.87%	6.37%	1.37%	4.39%	3.80%	4.63%	3.11%	(0.73)%
13	> 6-years and <= 7-years	6.5	3.78%	6.28%	1.28%	4.48%	3.53%	4.37%	3.19%	(0.68)%
14	> 7-years and <= 8-years	7.5	3.72%	6.22%	1.22%	4.56%	3.33%	4.18%	3.26%	(0.63)%
15	> 8-years and <= 9-years	8.5	3.68%	6.18%	1.18%	4.64%	3.17%	4.04%	3.32%	(0.58)%
16	> 9-years and <= 10-years	9.5	3.65%	6.15%	1.15%	4.69%	3.06%	3.93%	3.37%	(0.53)%
17	> 10-years and <= 15-years	12.5	3.63%	6.13%	1.13%	4.84%	2.88%	3.76%	3.50%	(0.38)%
18	> 15-years and <= 20-years	17.5	3.58%	6.08%	1.08%	4.89%	2.72%	3.62%	3.54%	(0.13)%
19	20-years	25	3.56%	6.06%	1.06%	4.90%	2.67%	3.57%	3.55%	0.00%



All shocks are currency-specific. This section also includes the interest rate floor specified in the PRA Rules, which is applied in the event shifted yield curves extend below this pre-determined level (red-dashed line on chart).

4.5 Inputs

This section provides details of the various inputs used in the formation of IRRBB results. This includes currency-level inputs including yield curve, NMD behavioural allocations (if any), and other behavioural adjustments (if applied).

Report date	28-Feb-23
Populate these cells	

Bucket No	Time Bucket	Mid-Point (Yrs)	0 Current Rates
1	Overnight	0.0028	3.93%
2	> Overnight and <= 1-month	0.0417	3.93%
3	> 1-month and <= 3-months	0.1667	4.10%
4	> 3-months and <= 6-momths	0.375	4.25%
5	> 6-months and <= 9-months	0.625	4.39%
6	> 9-months and <= 1-year	0.875	4.47%
7	> 1-year and <= 1.5-years	1.25	4.49%
8	> 1.5-years and <= 2-years	1.75	4.44%
9	> 2-years and <= 3-years	2.5	4.29%
10	> 3-years and <= 4-years	3.5	4.12%
11	> 4-years and <= 5-years	4.5	3.99%
12	> 5-years and <= 6-years	5.5	3.87%
13	> 6-years and <= 7-years	6.5	3.78%
14	> 7-years and <= 8-years	7.5	3.72%
15	> 8-years and <= 9-years	8.5	3.68%
16	> 9-years and <= 10-years	9.5	3.65%
17	> 10-years and <= 15-years	12.5	3.63%
18	> 15-years and <= 20-years	17.5	3.58%
19	20-years	25	3.56%

Behaviourally Affected Balance Sheet It Category 2: Non-Maturing Deposits	Core Deposits	Non-Core Deposits	Weighted Av. Tenor (Yr) Core	Overnight 0.0028	> Overnight and <= 1-month 0.0417	> 1-month 3-
2.1 Retail deposits in transactional accounts						
2.1.1 Current Account	90.00%	10.00%	3.00	0.00%	1.39%	2.78
Portfolio 2.1.2		100.00%	0.00			
Portfolio 2.1.3		100.00%	0.00			
Portfolio 2.1.4		100.00%	0.00			
Portfolio 2.1.5		100.00%	0.00			
Portfolio 2.1.6		100.00%	0.00			
Portfolio 2.1.7		100.00%	0.00			
Portfolio 2.1.8		100.00%	0.00			
Portfolio 2.1.9		100.00%	0.00			
Portfolio 2.1.10		100.00%	0.00			
Portfolio 2.1.11		100.00%	0.00			
Portfolio 2.1.12		100.00%	0.00			
Portfolio 2.1.13		100.00%	0.00			
Portfolio 2.1.14		100.00%	0.00			
Portfolio 2.1.15		100.00%	0.00			
2.2 Retail deposits in non-transactional, non-interest/profit bearing accounts						
2.2.1 Holding Account	8.70%	91.30%	3.00	0.00%	1.39%	2.78
Portfolio 2.2.2		100.00%	0.00			

Input (GBP)

Furthermore, it includes a summarised datamaster sheet as produced by the k-ALM IRRBB Rules Engine based on the data input and parameterisation specified during the implementation.

bs_category	currency_coc	row_ic	ovr_bucket_i	amount_rcy	amount_rcy_it
ASSET	GBP	A5.1	2	11,080,420	11,080
ASSET	GBP	A5.2	10	78,000,000	78,000
ASSET	GBP	A5.2	2	345,506,870	345,507
ASSET	GBP	A5.3	2	110,935,560	110,936
ASSET	GBP	A5.5	1	370,620	371
ASSET	GBP	A5.5	10	171,332,070	171,332
ASSET	GBP	A5.5	11	234,369,710	234,370
ASSET	GBP	A5.5	12	880,790	881

PRRBB_DATAMASTER_SUMMARY

5 Appendices

5.1 Appendix A – EvE Standardised Framework

The SF is set out in the ICAA Rules, 9.13 to 9.43. Below is our high-level guide on applying the SF.

PRA's Standardised Framework		
Step	**Action**	**Further recommendation**
Material currency identification		
1	Identify material currencies, defined as currencies in which the firm has banking book assets (or liabilities) of more than 5% of total assets (or liabilities).[13]	n/a
Balance sheet allocation		
2	Assign, in each material currency: 1) assets (excluding assets deducted from common equity tier 1 capital (CET1), fixed and intangible assets, and banking book equity exposures); 2) liabilities (excluding CET1 capital); and 3) off-balance sheet items, to the following categories: - (1) automatic interest rate options; [14] - (2) NMD; - (3) retail fixed-rate loans with prepayment risk; - (4) retail term deposits with early termination risk; and, - (5) other positions.	n/a
Sub-allocation: non-maturity deposits		
3A	Assign, in each material currency, NMD as follows: - (1) Retail deposits in non-interest or transactional accounts;[15] - (2) Other deposits from individuals; and, - (3) Other deposits. For each sub-category above, allocate the NMD between: - (A) Core, which means deposits that are found to remain undrawn with a high degree of likelihood using data history of an appropriate length, and unlikely to re-price even under significant changes in the interest rate environment; and, - (B) Non-core, which means any deposit that is not a core deposit. The following limitations apply to the Core elements of each sub-category as identified from 1-3 and A-B above: - 1 with A: 90% of balances, maximum re-pricing 5-years.	NMD from financial wholesale counterparties shall be immediately classified as non-core (see SS31/15, 2.9B(vii)). For NMD that are subject to periods of notice for repricing, we suggest allocation of non-core balances to the minimum period at which repricing can arise – in some cases this will be overnight, in others it may be the notice period (e.g., 35-days). The bank should document this assumption as part of its IRRBB Framework.

[13] If the material currencies identified capture less than 90% of total banking book assets (or liabilities), the bank shall include additional currencies as material currencies until the aggregated assets (or liabilities) of significant currencies equates to at least 90% of banking (non-trading) book assets (or liabilities).
[14] Which include *inter alia*: option derivative positions, term deposits by wholesale depositors that can be terminated early with no or an insignificant penalty; wholesale fixed rate loans subject to prepayment risk; and, mortgage loans with embedded caps or floors.
[15] Deposits from small businesses that the bank manages as retail exposures **may** be treated as retail deposits for these purposes, provided the customers' total aggregated liabilities are less than £877,000.

<table>
<tr><td></td><td>- 2 with A: 70% of balances, maximum re-pricing 4.5-years.
- 3 with A: 50% of balances, maximum re-pricing 4-years.

NMD assigned to the following categories shall be allocated to re-pricing time buckets (see Step 5) as follows:
- 1, 2 and 3 with A: estimate of re-pricing time bucket; and,
- 1, 2 and 3 with B: overnight time bucket.</td><td></td></tr>
<tr><td colspan="3">Sub-allocation: retail fixed rate loans with prepayment risk</td></tr>
<tr><td>3B</td><td>Assign, in each material currency, retail fixed-rate loans with prepayment risk to portfolios of homogenous positions.

For each portfolio, and in each material currency, determine the baseline conditional monthly prepayment rate (CMPR). Apply the following multiplier to the baseline CMPR of each portfolio and in each material currency to determine the CMPR for each shock scenario.

<table><tr><th>Scenarios</th><th>Multiplier</th></tr><tr><td>1, 3, 5</td><td>0.8</td></tr><tr><td>2, 4, 6</td><td>1.2</td></tr></table>
For each portfolio, and in each material currency, positions shall be allocated to re-pricing time buckets (see Step 5) as follows. (Refer to Step 4 for re-pricing cash flows.)

- <u>Overnight</u>: any positions with overnight re-pricing, plus 5% of the conditionally repaid amount (which is the scaled CMPR multiplied by the notional outstanding before repayments).
- <u>All other time buckets</u>: the lower of:
1) The re-pricing cash flows for all buckets less the re-pricing cash flows for all buckets from 1 (overnight) to the preceding time-bucket; and,
2) The baseline re-pricing cash flows for the bucket plus the number of months in the bucket multiplied by the scaled CMPR (the product of which shall be capped at 100%) multiplied by the notional after re-pricing cash flows in the preceding time-bucket have transpired.</td><td>We suggest “homogenous portfolios” be set out in the bank’s IRRBB Framework, which might define this as assets of a similar nature (which might further reference any of product type, customer type, geography, sector etc.)

We suggest banks calculate the baseline CMPR using a data-driven approach and based on recent historical observations. In times of significant change in the interest rate environment, banks might find it useful to apply management overlays, which should be reasonable and subject to appropriate governance.</td></tr>
</table>

Continued…

<table>
<tr><th colspan="3">Sub-allocation: retail term deposits with early termination risk</th></tr>
<tr><td>3C</td><td>Assign, in each material currency, retail term deposits with early termination risk to portfolios of homogenous positions.

For each portfolio, and in each material currency, determine the baseline term deposit redemption ratio (TDRR). Apply the following multiplier to the baseline TDRR of each portfolio and in each material currency to determine the TDRR for each shock scenario.

<table><tr><th>Scenarios</th><th>Multiplier</th></tr><tr><td>1, 4, 5</td><td>1.2</td></tr><tr><td>2, 3, 6</td><td>0.8</td></tr></table>
For each portfolio, and in each material currency, positions shall be allocated to re-pricing time buckets (see Step 5) as follows. Refer to Step 4 for re-pricing cash flows.

- <u>Overnight</u>: any positions with overnight re-pricing plus the conditionally redeemed amount (which is the scaled TDRR multiplied by the total term deposits subject to early redemption).
- <u>All other time buckets</u>: the baseline re-pricing cash flows for the re-pricing bucket multiplied by 1 minus the scaled TDRR.</td><td>See 3B for “homogenous positions”.

We suggest banks calculate the baseline TDRR using a data-driven approach and based on recent historical observations. In times of significant change in the interest rate environment, banks might find it useful to apply management overlays, which should be reasonable and subject to appropriate governance.</td></tr>
<tr><th colspan="3">Quantify re-pricing cash flows</th></tr>
<tr><td>4</td><td>For each category and sub-allocation, and in each material currency, determine the re-pricing cashflows as follows:
- repayments of principal amounts;
- re-pricing of principal amounts; and,
- interest payments.</td><td>For any floating rate instruments apply the coupon payments up until the next re-pricing date, and the notional amount (next re-pricing date).</td></tr>
</table>

Continued…

Re-pricing time bucket allocation		
5	Allocate each of the re-pricing cashflows, for each category and sub-allocation, and in each material currency, to the following time-buckets based on the re-pricing date of the cashflow.	As of 7 September 2023, note that there is an error in the ICAA Rules, which refers to bucket 3 mid-point as 0.0147-years rather than 0.0417-years.

Bucket	Interval	Midpoint (Years)
1	Overnight	0.0028
2	> Overnight and <= 1-month	0.0417
3	> 1-month and <= 3-months	0.1667
4	> 3-months and <= 6-momths	0.375
5	> 6-months and <= 9-months	0.625
6	> 9-months and <= 1-year	0.875
7	> 1-year and <= 1.5-years	1.25
8	> 1.5-years and <= 2-years	1.75
9	> 2-years and <= 3-years	2.5
10	> 3-years and <= 4-years	3.5
11	> 4-years and <= 5-years	4.5
12	> 5-years and <= 6-years	5.5
13	> 6-years and <= 7-years	6.5
14	> 7-years and <= 8-years	7.5
15	> 8-years and <= 9-years	8.5
16	> 9-years and <= 10-years	9.5
17	> 10-years and <= 15-years	12.5
18	> 15-years and <= 20-years	17.5
19	20-years	25

Risk free rates		
6	For each material currency, obtain the risk-free zero-coupon rate including any commercial margin and other spread components for each re-pricing time bucket.	A bank may elect to exclude commercial margin and other spread components where it is determined that the effect of their removal is immaterial, or if the bank has a reasonable process for deducting these elements. We suggest banks give strong consideration to the question of materiality on the basis of straightforwardness and proportionality. A formal definition of risk-free rate is lacking and there is no universal risk-free spot curve per currency. That being said, the EBA has stated that:[16]

[16] EBA/GL/2018/02, 115(n)

<table>
<tr><td></td><td></td><td colspan="2">

"An appropriate general 'risk-free' yield curve per currency should be applied (e.g. swap rate curves). That curve should not include instrument-specific or entity-specific credit spreads or liquidity spreads."

More recently, in the final draft Regulatory Technical Standards on Supervisory Outlier Tests adopted by the EBA and submitted to the European Commission, the EBA confirmed that:[17]

"…an appropriate general 'risk-free' yield curve per currency shall be applied (e.g., an OIS curve)."

In the final draft Regulatory Technical Standards on Standardised Approaches adopted by the EBA and submitted to the European Commission, the EBA suggested it was:[18] "left to institutions to select it [the risk free-curve]".

Furthermore, the BCBS stated:[19]

"An example of an acceptable yield curve is a secured interest rate swap curve."

Market data providers such as Bloomberg also provide functions or tools to obtain risk-free zero-coupon rates. The rate should be obtained for the bucket mid-point, i.e., for bucket 7 (> 1-year and <= 1.5-years) the rate should be obtained for the mid-point of 1.25-years.

</td></tr>
<tr><td colspan="4">Interest rate scenarios</td></tr>
<tr><td>7</td><td colspan="2">

For each material currency, establish the interest rate shift for each re-pricing time-bucket associated with the following interest rate shock scenarios. Refer also to **Appendix B** and **C**.

<table>
<tr><th>Scenario</th><th>Description</th><th>Requirement</th></tr>
<tr><td>1</td><td>Parallel up</td><td>As specified for the material currency (e.g., GBP 250bp).</td></tr>
<tr><td>2</td><td>Parallel down</td><td>As specified for the material currency (e.g., GBP 250bp).</td></tr>
<tr><td>3</td><td>Steepener</td><td>-65% of the absolute value of the short rate shift for the mid-point of the re-pricing time bucket (see Scenario 5) plus 90% of the absolute value of the long rate shift for the mid-point of the re-pricing time bucket (see below table).</td></tr>
<tr><td>4</td><td>Flattener</td><td>80% of the absolute value of the short rate shift for the mid-point of the re-pricing time bucket (see Scenario 5) less 60% of the absolute value of the long rate shift for the mid-point of the re-pricing time bucket (see below table).</td></tr>
<tr><td>5</td><td>Short rates up</td><td>The product of the amount specified for the material currency (e.g., GBP 300bp) and the mathematical constant that is the base of the natural</td></tr>
</table>

</td><td>A bank must apply at a minimum the six prescribed scenarios. We also recommend a bank gives consideration to additional interest rate shock scenarios based on the composition of its book and considering the materiality of IRRBB to its business.</td></tr>
</table>

17 EBA/RTS/2022/10, article 4(m)
18 EBA/RTS/2022/09, 19(a)
19 BCBS Standards: Interest rate risk in the banking book, para. 70(i)(c), footnote 12

	<table><tr><td></td><td></td><td>logarithm (*e*) raised to the negative power of the midpoint of the re-pricing time bucket (in years) divided by the constant *x*, which is set at 4.</td></tr><tr><td>6</td><td>Short rates down</td><td>Negative of the outcomes of Scenario 5.</td></tr></table>	
Adjusted risk-free rates		
8	For each material currency, obtain the adjusted risk-free zero-coupon rate by applying the interest rate shift at each midpoint of a re-pricing time bucket (see **Step 7**) to the baseline risk-free zero-coupon rate. The adjusted risk-free zero-coupon rate at the midpoint of a re-pricing time bucket shall be floored at -100bp for maturities up to 1-year. The floor shall rise by 5bp per year until the floor reaches 0bp for maturities of 20-years of more. If observed rates are lower than floor of -100 basis points, the bank should apply the lower observed rates.	n/a

Continued…

Discount factors		
9	For each material currency, assign a discount factor to the midpoint of each re-pricing time bucket and for each interest rate shock scenario. The discount factor shall be: - The mathematical constant that is the base of the natural logarithm (*e*) raised to the negative power of the adjusted risk-free zero-coupon rate for the midpoint of the re-pricing time bucket (see **Step 8**) multiplied by the midpoint of the re-pricing time bucket in years (see **Step 5**).	n/a
Change in economic value of equity (ΔEVE)		
10	For banks whose positions do not include automatic interest rate option risk (see **Step 1**), for each material currency, and for each interest rate shock scenario, the change in EVE (ΔEVE) is equal to the sum of the change in non-automatic interest rate option risk (NAO) for all re-pricing time-buckets.[20] For each material currency, and for each interest rate shock scenario, the change in NAO is the sum of the re-pricing cash flows allocated across re-pricing time-buckets in the baseline scenario multiplied by the base discount factor per bucket less the sum of the re-pricing cash flows allocated across the re-pricing time-buckets in the shock scenario multiplied by the adjusted discount factor per bucket (reflecting the adjusted risk-free zero-coupon rate for the scenario).	n/a
11	The bank shall perform the calculation in **Step 10** for each material currency and interest rate shock scenario. The bank shall aggregate the results for each interest rate shock scenario by combining the results in each material currency. The results in each material currency shall be combined on the following basis: - Each positive amount; plus, - 50% of each negative amount. The result is the ΔEVE on a consolidated currency basis for the interest rate shock scenario. The bank shall derive EVE_{LOSS} by taking the maximum ΔEVE across all scenarios.	Note that ΔEVE – under the PRA's approach – will be positive in loss scenarios, and negative in gain scenarios.
Supervisory Outlier Test		
12	The bank shall measure the result established in **Step 11** as a proportion of its Tier 1 capital. Where this value exceeds 15%, the bank must immediately notify the PRA.	See ICAA Rules, 9.4A.

5.2 Appendix B – Interest Rate Shifts

Sub-set of currency-specific interest rate shifts.

Currency	Parallel	Short	Long
EUR	200	250	100
GBP	250	300	150
USD	200	300	150

[20] For banks with automatic interest rate option risk, it must add to the change in NAO the change in value of the automatic interest rate options (KAO).

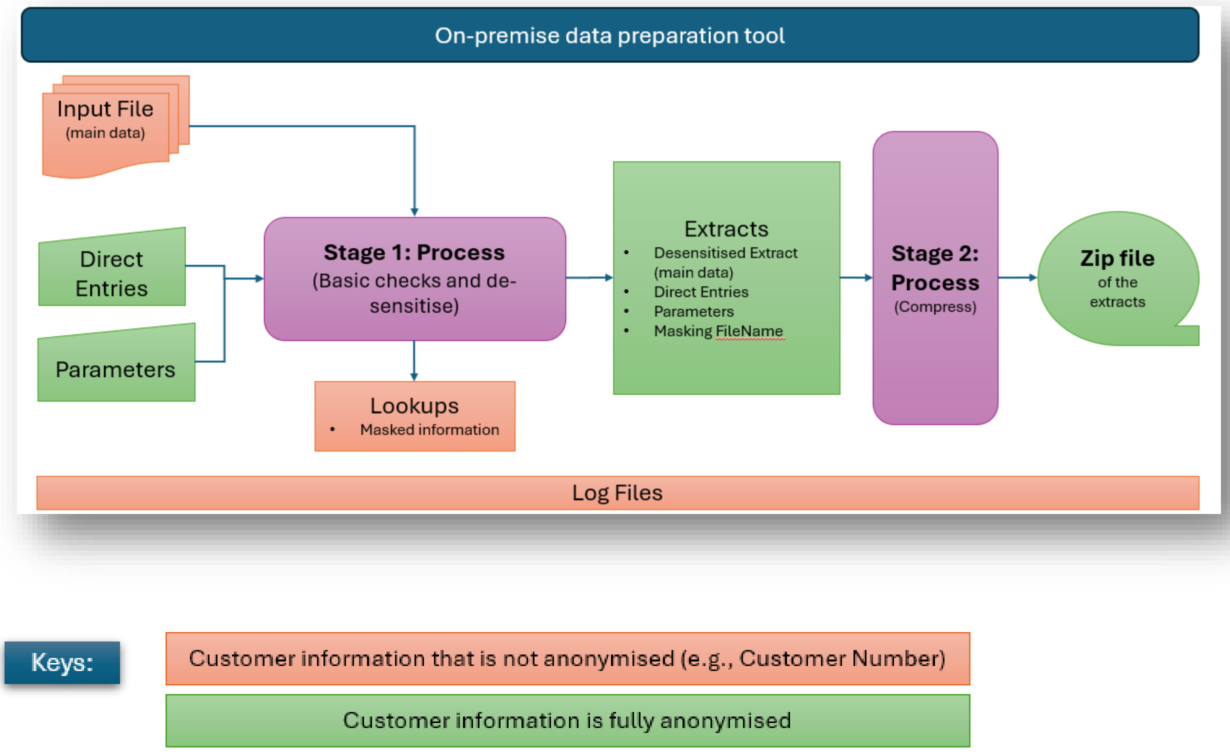
5.3 Appendix C – Rate Shift Formulas

Interest rate shift formulas. Refer to Appendix B for currency-specific inputs.

Scenario	Formula
1	+ Parallel
2	- Parallel
3	-0.65 x Short shift + 0.9 x Long shift
4	0.8 x Short shift – 0.6 x Long shift
5	+ Short shift
6	- Short shift
Short shift	Short x $e^{-t/x}$
Long shift	Long x (1 – $e^{-t/x}$)

And where: x = 4; e is the base of the natural logarithm.

5.4 Appendix D – The DataPrep Tool process flow



- Input file is the main data file with transaction level information in prescribed format BankName_kALM_I/PRRBB_RawData_DYYYYMMDD.xlsx XYZUK_kALM_IRRBB_RawData_D20240930.xlsx)
- The Zip file created is the desensitized/anonymised file to be uploaded into the k-ALM cloud secure environment
- The Lookups folder/files contains the mapping for the given reporting reference date of the actual customer number and contract reference number to the dummy customer/contract numbers created as part of the anonymisation process
- Log files provide additional information for each run, which might be helpful in analysing the overall process

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